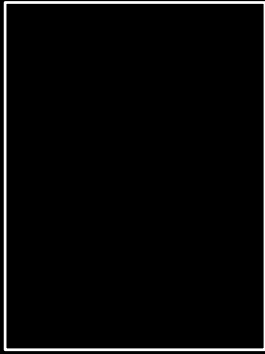


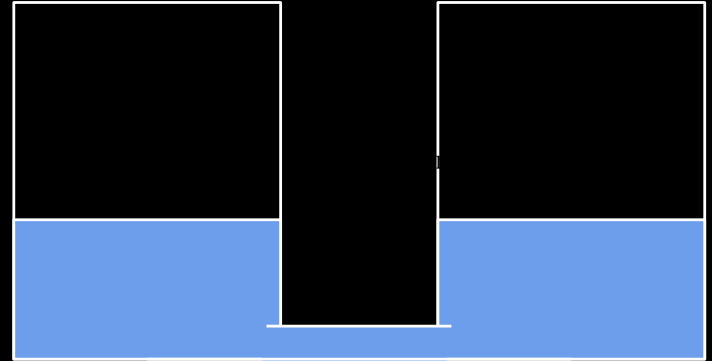
# Chemical Equilibrium



A



B



Equilibrium

A

B

Water flow stops when water levels  
becomes equal in both A to B

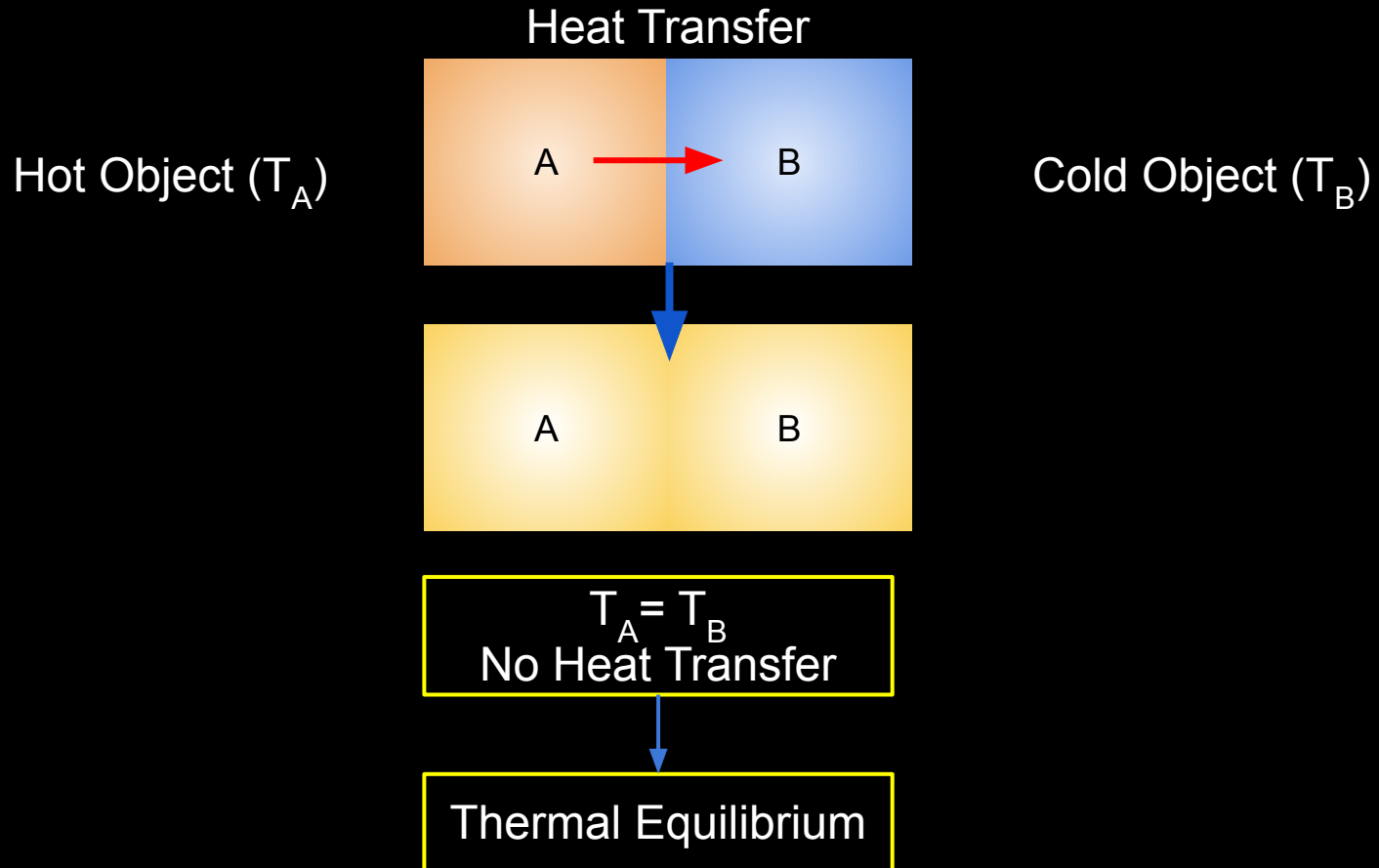


$$F_A = F_B$$

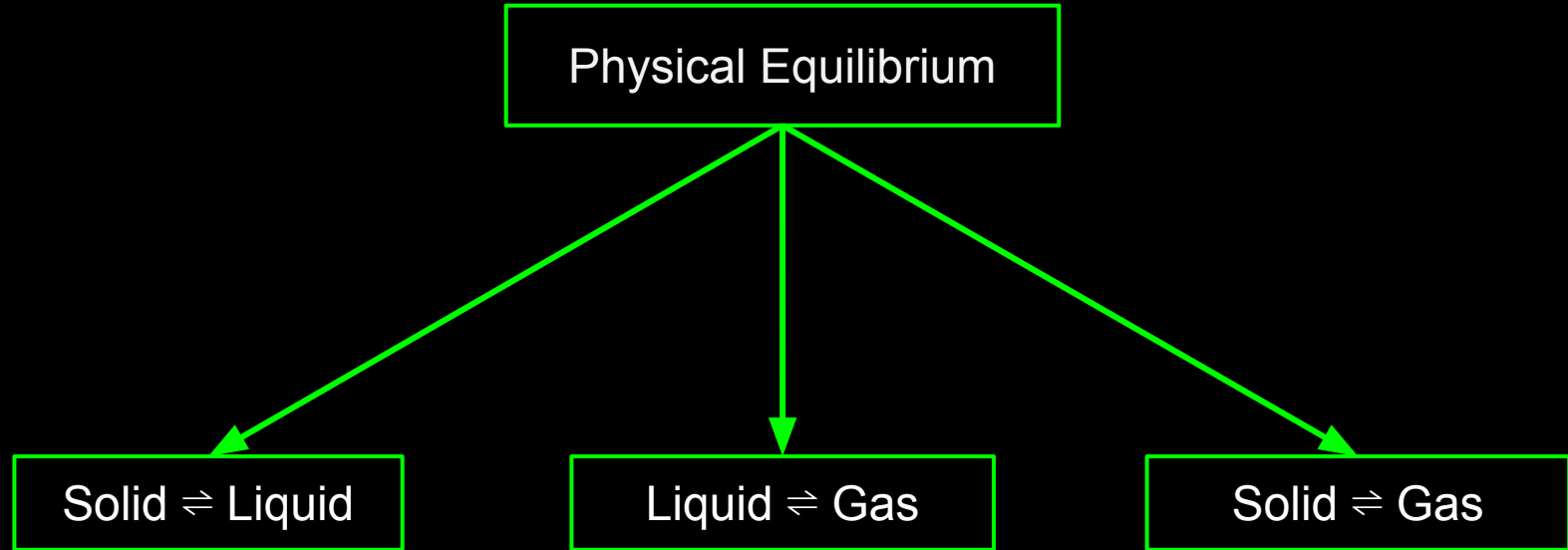
No displacement of block



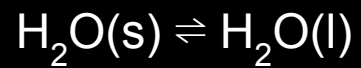
Mechanical Equilibrium



# Equilibrium for different Physical Processes



# Solid-Liquid Equilibrium

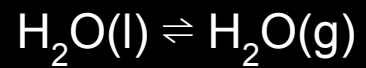


Melting Point

rate of melting = rate of freezing

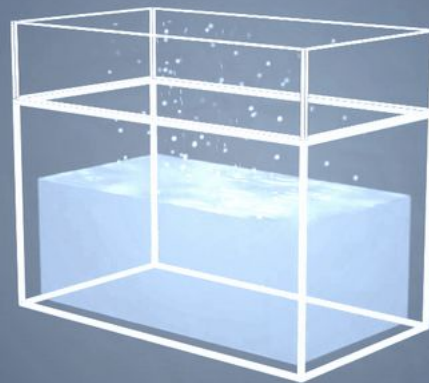


# Liquid-Vapour Equilibrium

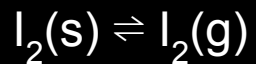


Boiling Point

rate of evaporation = rate of condensation

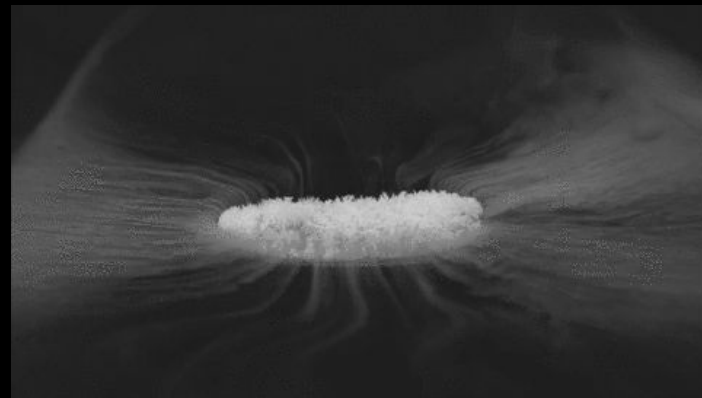


# Solid-Gas Equilibrium



Sublimation Temperature

rate of sublimation = rate of deposition



# General Characteristics of Equilibria Involving Physical Processes

1. Possible only in a closed system at a given temperature
2. Both the processes occur at the same rate and there is a dynamic equilibrium
3. All measurable properties of the system remain constant  Vapour pressure  
Boiling Point  
Freezing Point
4. Characterised by constant value of one of its parameters at a given temperature
5. The magnitude of such quantities at any stage indicates the extent to which the physical process has proceeded before reaching equilibrium

# Solubility Equilibria

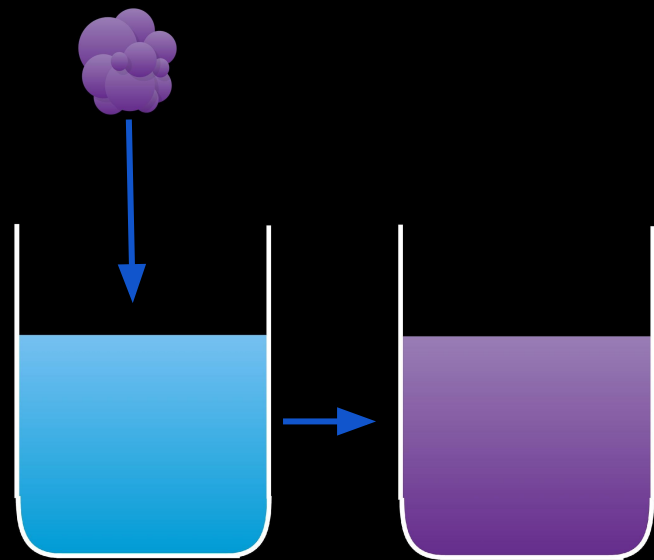
## Solid in Liquid:



No more solid can be dissolved

Saturated Solution

rate of dissolution = rate of crystallisation

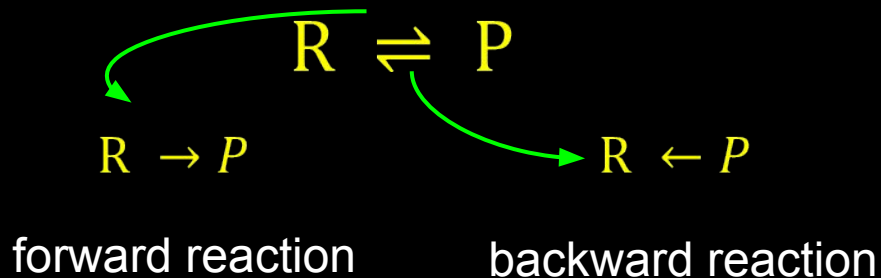


## Solubility:

Maximum amount of solute dissolved in solution at saturated condition

|                                 |                         |                                       |
|---------------------------------|-------------------------|---------------------------------------|
| Amount dissolved $<$ Solubility | Unsaturated Solution    | More solute can be dissolved          |
| Amount dissolved $=$ Solubility | Saturated Solution      | No more solute can be dissolved       |
| Amount dissolved $>$ Solubility | Supersaturated Solution | Solute crystallizes and separated out |

# Equilibrium in Chemical Processes



Rate of forward reaction = rate of backward reaction

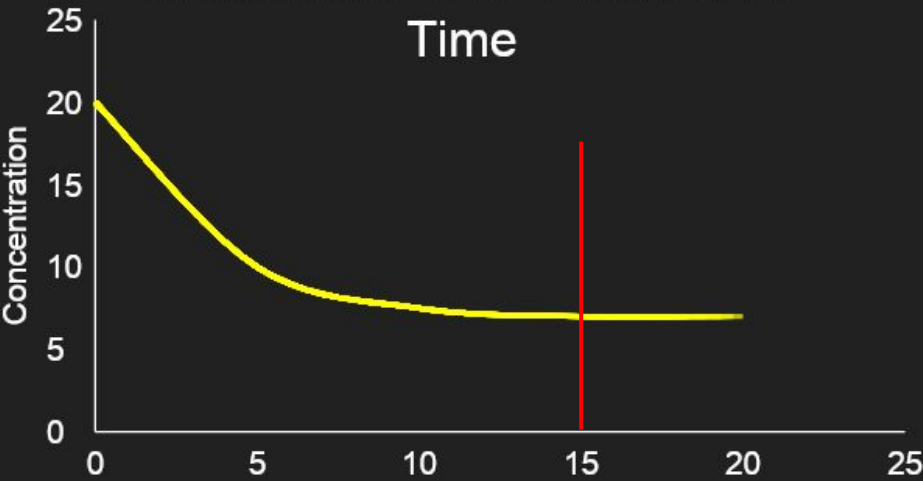
No change in concentrations of reactants and products

Chemical Equilibrium

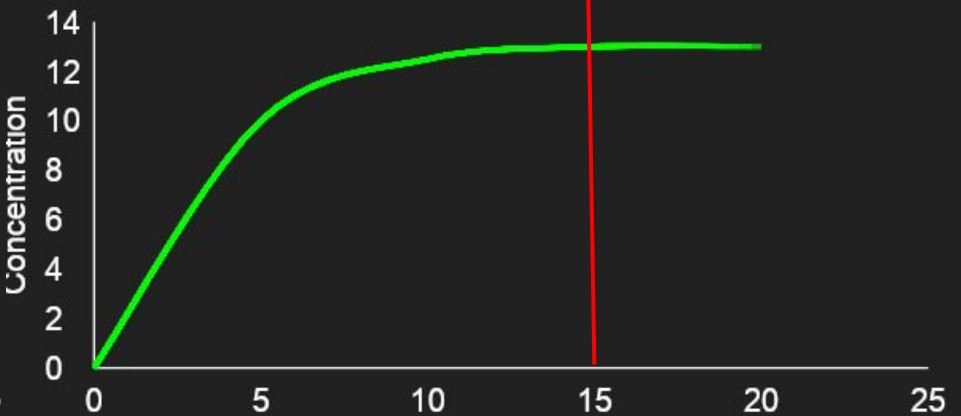
Dynamic Equilibrium

# Equilibrium in Chemical Processes

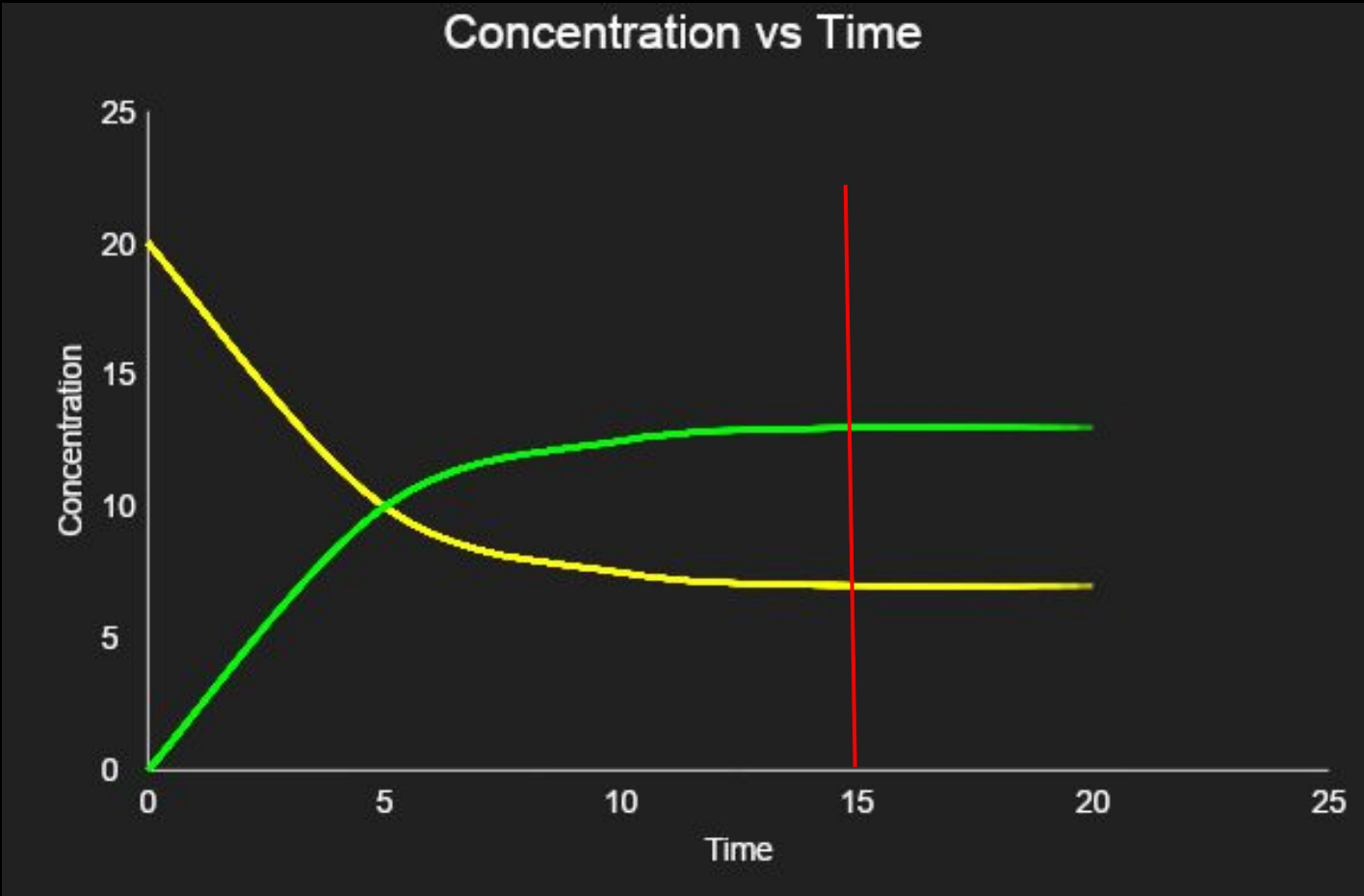
Concentration of Reactant vs Time



Concentration of Product vs Time

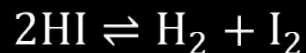
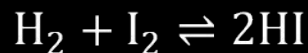


# Equilibrium in Chemical Processes

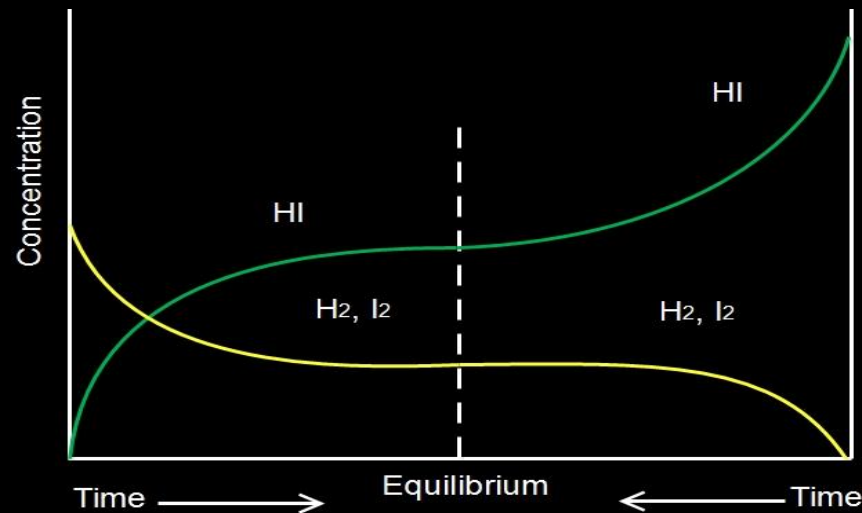


# Characteristics of Chemical Equilibrium

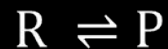
1. Possible only in closed system
2. Dynamic in nature
3. Properties of system remain constant
4. Chemical equilibrium can be achieved from the both sides



Temperature  
Pressure  
Colour  
Concentration



# Law of Mass Action



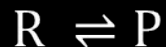
For any reaction,

Rate of reaction  $\propto$  Active mass of reactants

**Active mass:** Mass of substance participating in a chemical reaction

|             |                     |
|-------------|---------------------|
| Solution    | Molar concentration |
| Gases       | Partial pressure    |
| Pure solid  | Constant (=1)       |
| Pure Liquid | Constant (=1)       |

# Law of Mass Action



Rate of reaction  $\propto$  Active mass of reactants

Rate of forward reaction( $r_f$ )  $\propto$  [R]

$$r_f = k_f[R]$$

$k_f$  is rate constant of forward reaction

Rate of backward reaction( $r_b$ )  $\propto$  [P]

$$r_b = k_b[P]$$

$k_b$  is rate constant of backward reaction

At equilibrium,

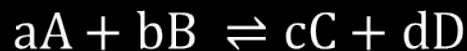
Rate of forward reaction = Rate of backward reaction

$$r_f = r_b$$

$$k_f[R] = k_b[P]$$

$$\frac{k_f}{k_b} = \frac{[P]}{[R]} = K_{eq} = \text{Equilibrium constant}$$

# Law of Chemical Equilibrium



$$r_f = k_f[A]^a[B]^b$$

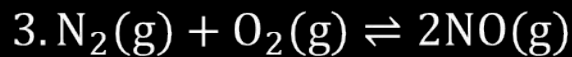
$$r_b = k_b[C]^c[D]^d$$

$$K_{eq} = \frac{k_f}{k_b} = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

Law of Chemical Equilibrium

$$\text{Unit of } K_{eq} = \frac{[\text{molL}^{-1}]^c[\text{molL}^{-1}]^d}{[\text{molL}^{-1}]^a[\text{molL}^{-1}]^b} = [\text{molL}^{-1}]^{(c+d)-(a+b)}$$

Q. Write the expressions for  $K_{eq}$  for following reactions:



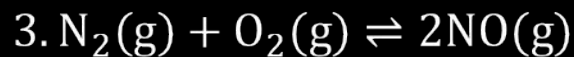
Sol.



$$K_{\text{eq}} = \frac{[\text{N}_2][\text{H}_2]^3}{[\text{NH}_3]^2}$$



$$K_{\text{eq}} = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]}$$



$$K_{\text{eq}} = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]}$$

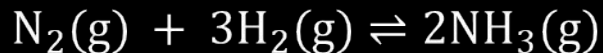


$$K_{\text{eq}} = [\text{CO}_2]$$

# Types of Equilibrium

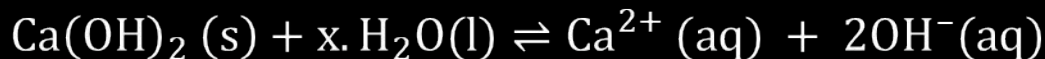
## Homogeneous Equilibria:

All the reactants and products are in the same phase



## Heterogeneous Equilibria:

The reactants and products are in more than one phase



## Summary

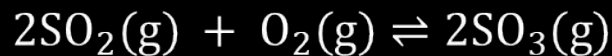
- Solubility of gas in liquid decreases with increase in temperature.
- Solubility of gas in liquid increases with increase in pressure
- Henry's Law:  $P = k_H \times \chi_{\text{gas in liquid}}$
- At equilibrium: Rate of forward reaction = rate of backward reaction
- For equilibrium:  $aA + bB \rightleftharpoons cC + dD$

$$K_{\text{eq}} = \frac{k_f}{k_b} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

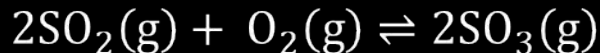
- Homogeneous Equilibria: All the reactants and products are in the same phase.
- Heterogeneous Equilibria: The reactants and products are in more than one phase.

**Q.** What is  $K_{eq}$  for the following equilibrium when the equilibrium concentration of each substance is:

$[SO_2] = 0.60M$ ,  $[O_2] = 0.82M$  and  $[SO_3] = 1.90M$  ?



**Sol.**

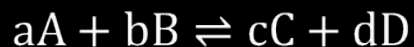
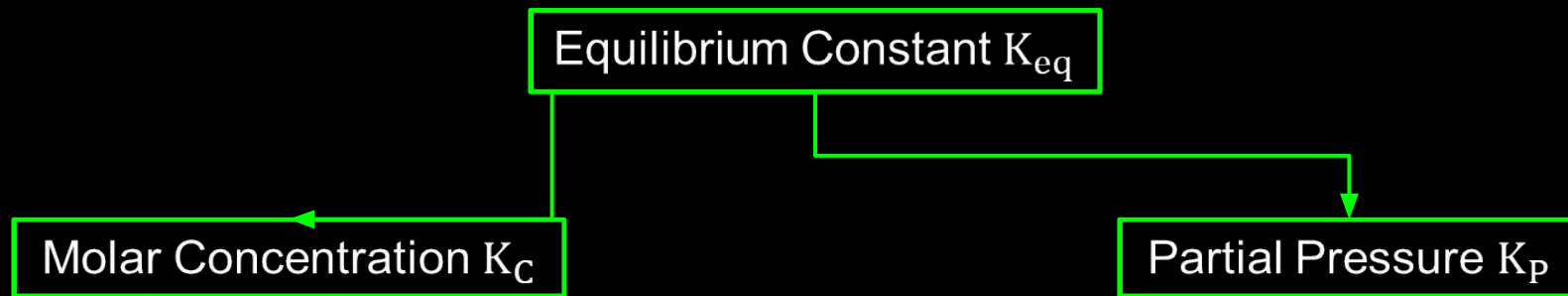


$$K_{eq} = \frac{[SO_3]^2}{[SO_2]^2[O_2]}$$

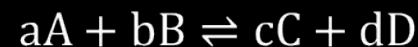
$$K_{eq} = \frac{1.90^2}{0.60^2 \times 0.82}$$

$$= 12.23 \text{ Lmol}^{-1}$$

# Types of Equilibrium Constant

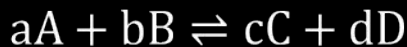


$$K_C = \frac{C_C^c \times C_D^d}{C_A^a \times C_B^b}$$



$$K_P = \frac{p_C^c \times p_D^d}{p_A^a \times p_B^b}$$

## Relation between $K_C$ and $K_P$



$$K_P = \frac{p_C^c \times p_D^d}{p_A^a \times p_B^b}$$

Putting values of  $P_A, P_B, P_C$  and  $P_D$

$$K_P = \frac{(C_C RT)^c \times (C_D RT)^d}{(C_A RT)^a \times (C_B RT)^b}$$

$$K_P = \frac{C_C^c \times C_D^d}{C_A^a \times C_B^b} \times (RT)^{(c+d)-(a+b)}$$

$$K_P = K_C \times (RT)^{\Delta n_g}$$

Similarly,

$$PV = nRT$$

$$P = \frac{n}{V} RT \quad \dots \frac{n}{V} = C$$

$$P = CRT$$

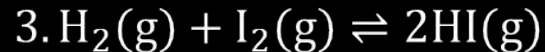
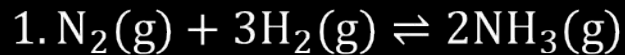
$$P_A = C_A RT$$

$$P_B = C_B RT$$

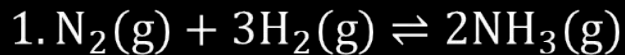
$$P_C = C_C RT$$

$$P_D = C_D RT$$

**Q.** Write the expressions showing the relation between  $K_C$  and  $K_P$  in following reactions:



Sol.



$$\Delta n_g = \Sigma n_g(\text{products}) - \Sigma n_g(\text{reactants})$$

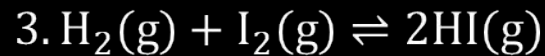
$$\Delta n_g = 2 - 4 = -2$$

$$K_P = K_C \times (RT)^{-2}$$



$$\Delta n_g = 2 - 1 = 1$$

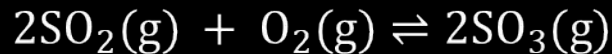
$$K_P = K_C \times (RT)$$



$$\Delta n_g = 2 - 2 = 0$$

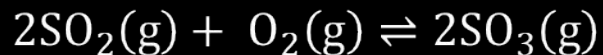
$$K_P = K_C$$

**Q.** At 450 K,  $K_P = 2.0 \times 10^{10} \text{ bar}^{-1}$  for the given reaction at equilibrium.



What is  $K_C$  at this temperature ?

**Sol.**



$$\Delta n_g = \Sigma n_g(\text{products}) + \Sigma n_g(\text{reactants})$$

$$\Delta n_g = 2 - 3 = -1$$

$$K_P = K_C \times (RT)^{\Delta n_g}$$

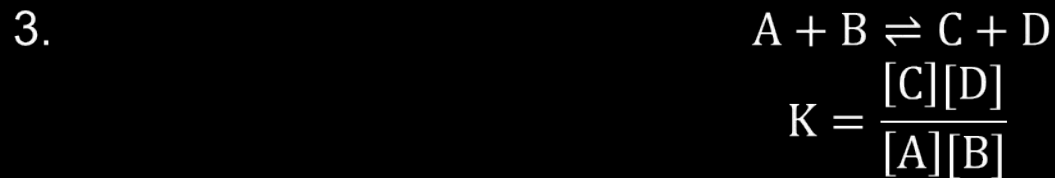
$$K_C = \frac{K_P}{(RT)^{-1}} = K_P \times RT$$

$$R = 0.0831 \text{ barLK}^{-1}\text{mol}^{-1}$$

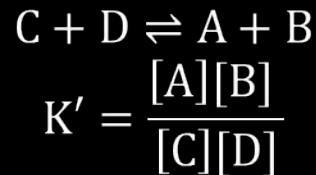
$$K_C = 2.0 \times 10^{10} \times 0.0831 \times 450 = 7.479 \times 10^{11} \text{ Lmol}^{-1}$$

# Properties of Equilibrium Constant

1. Equilibrium constant is reaction specific
2. For a given reaction  $K_{eq}$  depends only on temperature



If equation is reversed



then,

$$K' = \frac{1}{K}$$

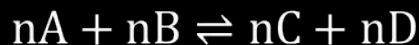
# Properties of Equilibrium Constant

4.



$$K = \frac{[C][D]}{[A][B]}$$

If equation is multiplied by a number 'n'



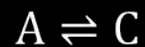
$$K' = \frac{[C]^n [D]^n}{[A]^n [B]^n} = \left[ \frac{[C][D]}{[A][B]} \right]^n$$

then,

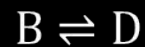
$$K' = K^n$$

## Properties of Equilibrium Constant

5.

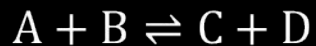


$$K_1 = \frac{[C]}{[A]} \quad \dots 1$$



$$K_2 = \frac{[D]}{[B]} \quad \dots 2$$

If we add both equations



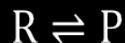
$$K' = \frac{[C][D]}{[A][B]} = \frac{[C]}{[A]} \times \frac{[D]}{[B]}$$

then,

$$K' = K_1 \times K_2$$

# Applications of Equilibrium Constant

## 1. Predict the extent of reaction



$$K_{eq} > 10^3$$

→ Large amount

$$K_{eq} = \frac{[P]}{[R]}$$

→ Negligible amount



Reaction proceeds almost  
to completion

$$K_{eq} < 10^{-3}$$

→ Negligible amount

$$K_{eq} = \frac{[P]}{[R]}$$

→ Large amount



Reaction hardly proceeds

$$10^{-3} < K_{eq} < 10^3$$

$$K_{eq} = \frac{[P]}{[R]}$$



Both reactant and product  
are at equilibrium

Q. In which of the following cases reaction goes fastest to completion? (K is equilibrium constant)

1.  $K = 10^{-2}$

2.  $K = 100$

3.  $K = 10$

4.  $K = 10^4$

**Q.** In which of the following cases reaction goes fastest to completion? (K is equilibrium constant)

1.  $K = 10^{-2}$
2.  $K = 100$
3.  $K = 10$
4.  $K = 10^4$

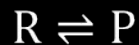
**Sol.** 4.  $K = 10^4$

Higher the value of K faster the rate of forward direction as

$$K = \frac{k_f}{k_b}$$

# Applications of Equilibrium Constant

## 2. Predict the direction of reaction



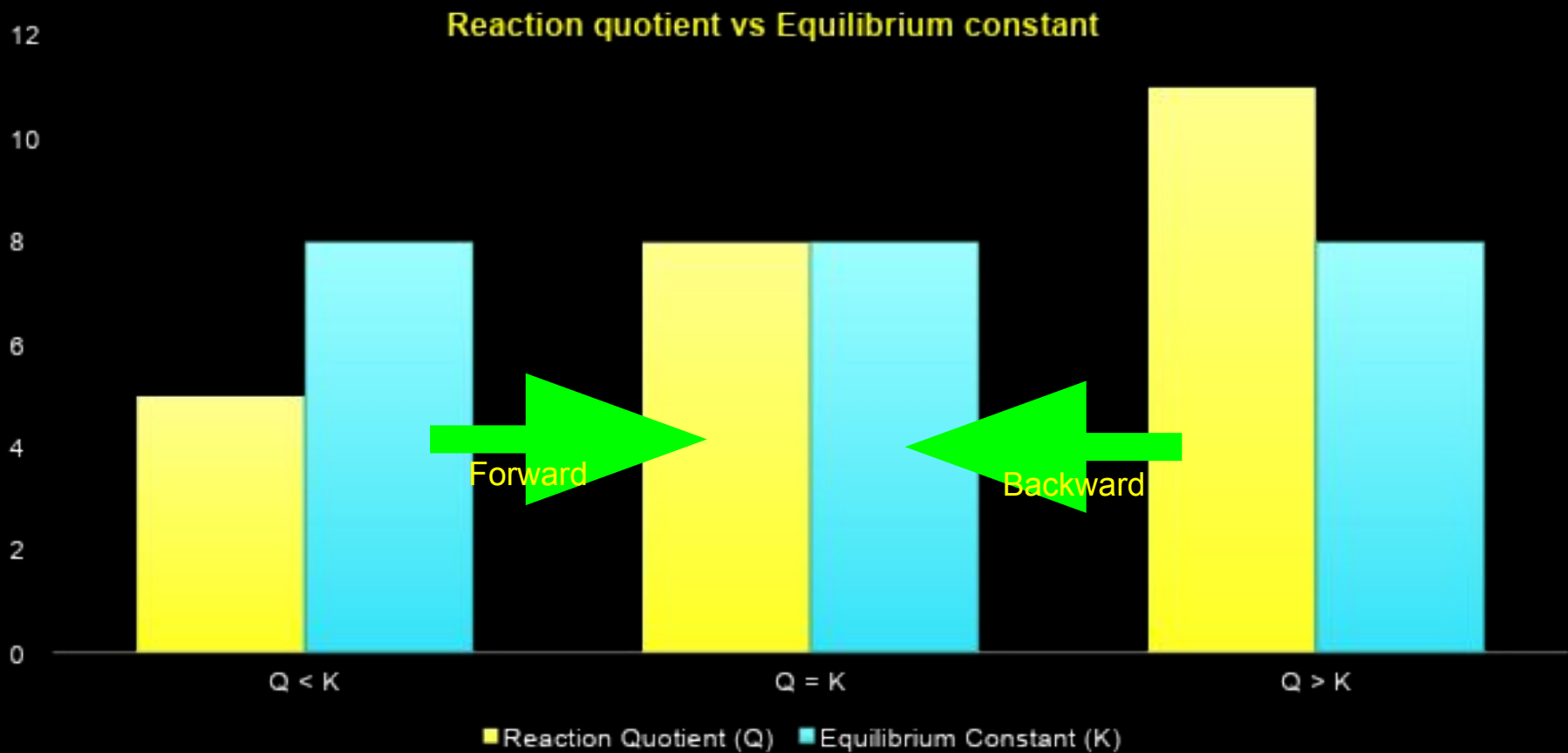
$$\text{Reaction Quotient}(Q) = \frac{[P]}{[R]}$$

Q is written at any instant of time where as K is written at equilibrium.

As we know every system tries to attain equilibrium so,

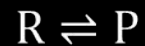
$$Q \rightarrow K$$

# Applications of Equilibrium Constant



# Applications of Equilibrium Constant

## 3. Calculation of Standard Gibbs Free Energy ( $\Delta_r G^0$ )



$$\Delta_r G^0 = -2.303 \times RT \log K_{eq}$$

**Q.** Equilibrium constant,  $K_c$  for the reaction



At a particular time, the analysis shows that composition of the reaction mixture is  $3.0 \text{ mol L}^{-1} \text{ N}_2$ ,  $2.0 \text{ mol L}^{-1} \text{ H}_2$  and  $0.5 \text{ mol L}^{-1} \text{ NH}_3$ . Is the reaction at equilibrium? If not in which direction does the reaction tend to proceed to reach equilibrium?

**Sol.**  $[\text{N}_2] = 3.0 \text{ mol L}^{-1}$ ;  $[\text{H}_2] = 2.0 \text{ mol L}^{-1}$ ;  $[\text{NH}_3] = 0.5 \text{ mol L}^{-1}$ ;  $K_{\text{eq}} = 0.061$

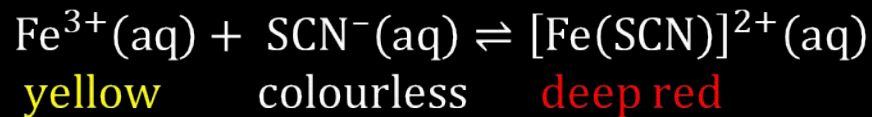
$$Q = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$
$$Q = \frac{(0.5)^2}{(3.0) \times (2.0)^3} = 0.0104 \text{ L}^2 \text{ mol}^{-2}$$

$Q < K_{\text{eq}} \Rightarrow$  forward direction

## Le-Chatelier's Principle

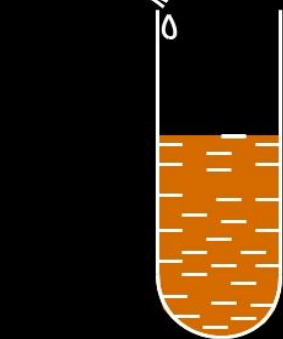
“A change in any of the factors that determine the equilibrium conditions of a system will cause the system to change in such a manner so as to reduce or to counteract the effect of the change”

# Changing the Concentrations of Reactants and Products



$$K_c = \frac{[[\text{Fe}(\text{SCN})]^{2+}]}{[\text{Fe}^{3+}][\text{SCN}^{-}]}$$

Addition of  
 $[\text{Fe}(\text{SCN})]^{2+}$

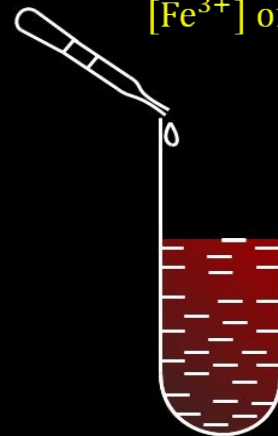


Backward  
Direction



Forward  
Direction

Addition of  
 $[\text{Fe}^{3+}]$  or  $[\text{SCN}^{-}]$



## Changing the Concentrations of Reactants and Products

| Change in concentration | Direction of Equilibrium Shift                                                      |
|-------------------------|-------------------------------------------------------------------------------------|
| Addition of Reactants   |  |
| Removal of Reactants    |  |
| Addition of Products    |  |
| Removal of Products     |  |

Q. Describe the effect of:

- a) Addition of  $\text{H}_2$
- b) Addition of  $\text{CH}_3\text{OH}$
- c) Removal of  $\text{CO}$
- d) Removal of  $\text{CH}_3\text{OH}$

on the equilibrium if the reaction:



Q. Describe the effect of:

- a) Addition of  $\text{H}_2$
- b) Addition of  $\text{CH}_3\text{OH}$
- c) Removal of  $\text{CO}$
- d) Removal of  $\text{CH}_3\text{OH}$

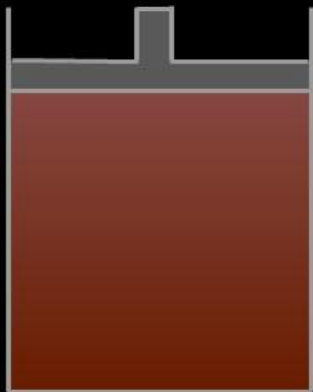
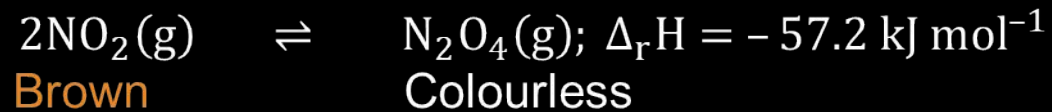
on the equilibrium if the reaction:



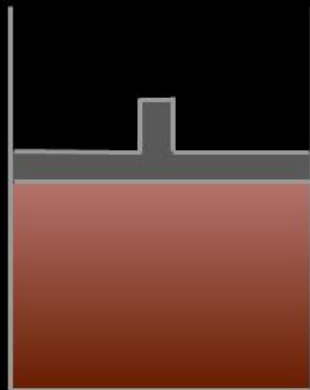
Sol.

- a) Shifts in forward direction
- b) Shifts in backward direction
- c) Shifts in backward direction
- d) Shifts in forward direction

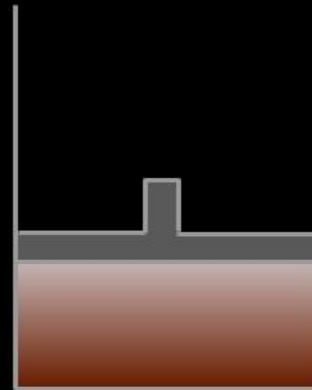
## Changing the Pressure



Backward  
Direction



Forward  
Direction



# Changing the Pressure

|  | Change in Pressure               | Direction of Equilibrium Shift                                                      |
|--|----------------------------------|-------------------------------------------------------------------------------------|
|  | Increase in Pressure             |  |
|  | Decrease in Pressure             |  |
|  | Increase in Pressure             |  |
|  | Decrease in Pressure             |  |
|  | Increase or decrease in Pressure | No shift                                                                            |

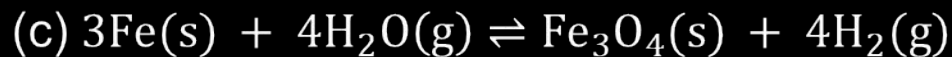
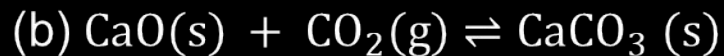
# Changing the Volume

At constant Temperature

$$V \propto \frac{1}{P}$$

|  | Change in Volume               | Direction of Equilibrium Shift                                                      |
|--|--------------------------------|-------------------------------------------------------------------------------------|
|  | Increase in Volume             |  |
|  | Decrease in Volume             |  |
|  | Increase in Volume             |  |
|  | Decrease in Volume             |  |
|  | Increase or decrease in Volume | No shift                                                                            |

**Q.** Does the number of moles of reaction products increase, decrease or remain same when each of the following equilibria is subjected to a decrease in pressure by increasing the volume?



**Sol.** Pressure of system decreases

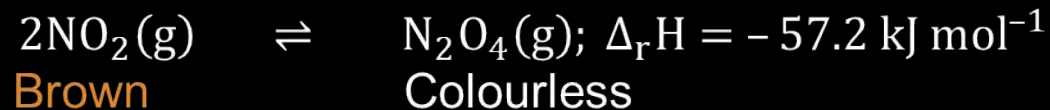
⇒ Equilibrium will shift towards the side with more number of **gaseous moles**.

(a) Forward Direction, moles of products will increase and reactants will decrease

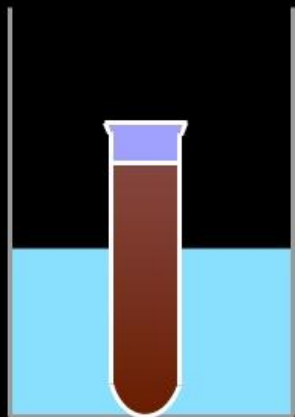
(b) Backward Direction, moles of reactants will increase and products will decrease

(c) No Shift, concentration remains same

# Changing the Temperature

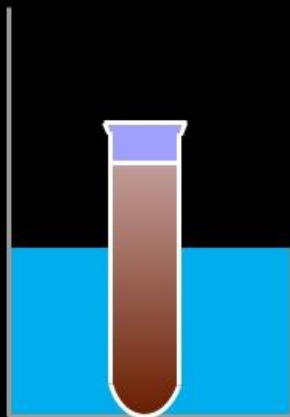


Water at 363 K



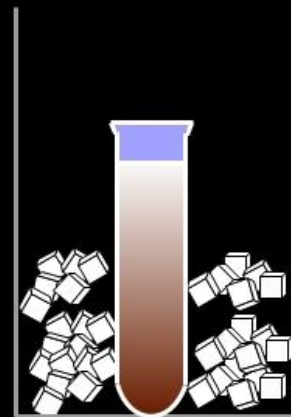
Backward  
Direction

Water at room  
temperature 270 K



Forward  
Direction

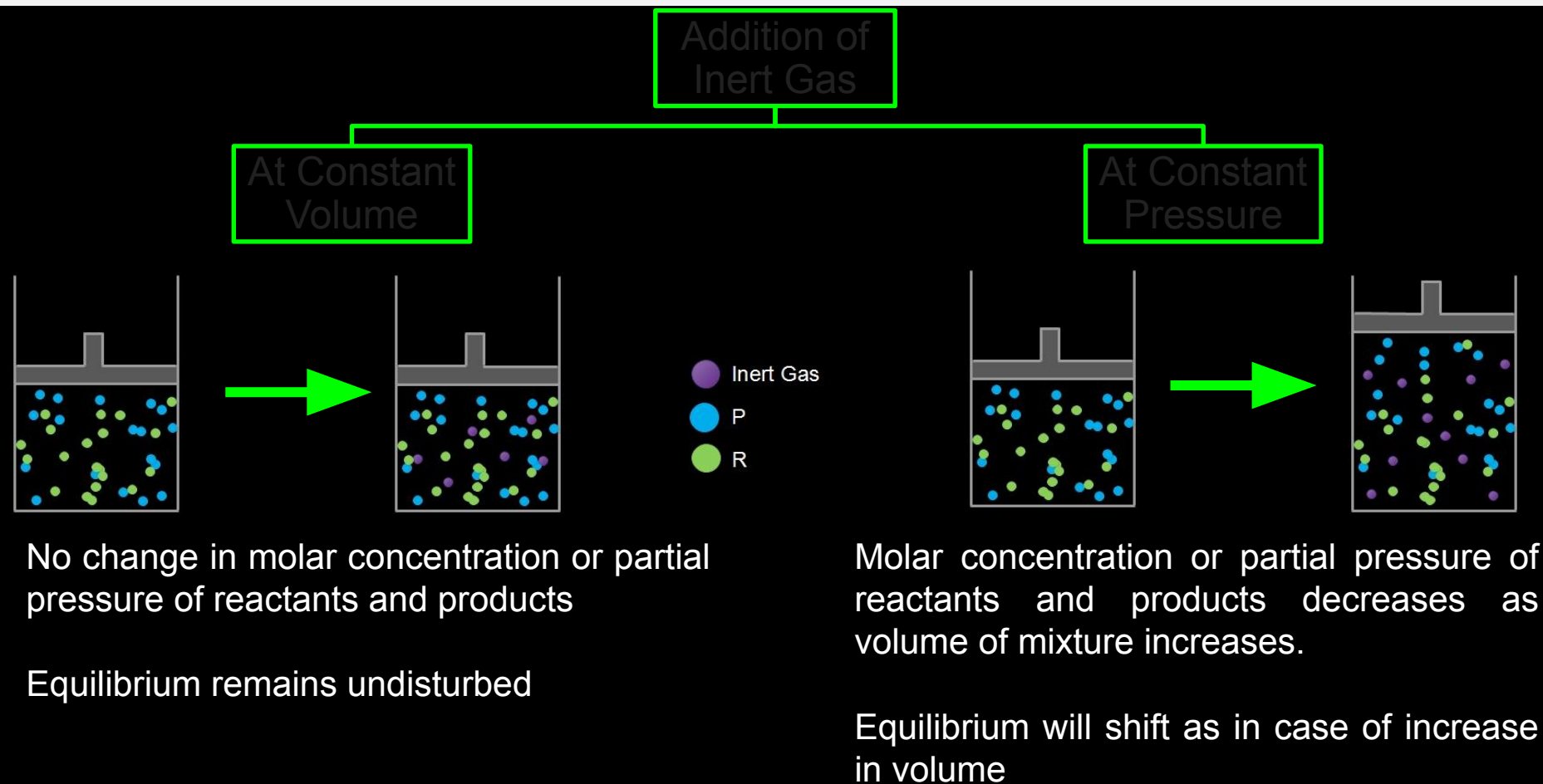
Freezing mixture  
at 270 K



## Changing the Temperature

|  | Change in Temperature   | Direction of Equilibrium Shift                                                      |
|--|-------------------------|-------------------------------------------------------------------------------------|
|  | Increase in Temperature |  |
|  | Decrease in Temperature |  |
|  | Increase in Temperature |  |
|  | Decrease in Temperature |  |

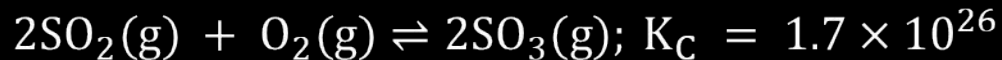
# Addition of Inert Gas



# Addition of Inert Gas

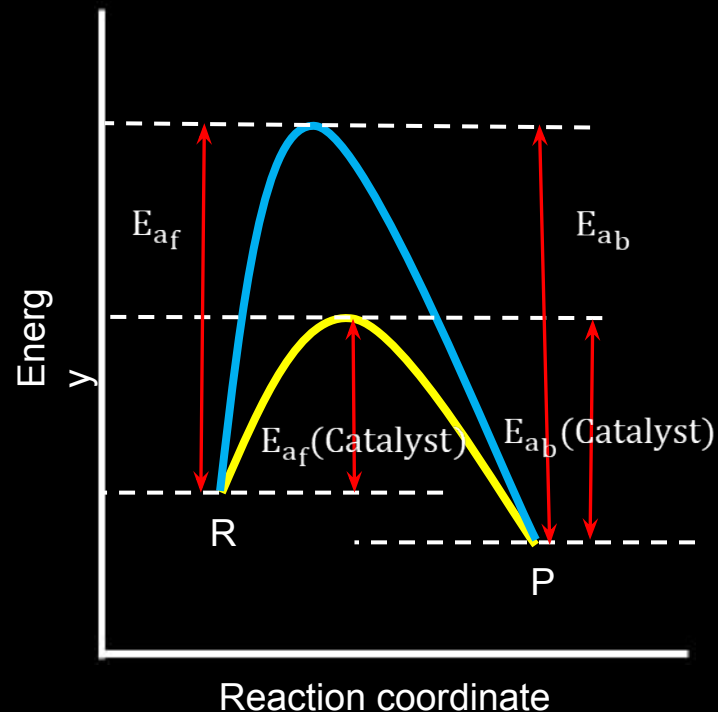
|  | Change in Volume               | Direction of Equilibrium Shift                                                      |
|--|--------------------------------|-------------------------------------------------------------------------------------|
|  | Increase in Volume             |  |
|  | Increase in Volume             |  |
|  | Increase or decrease in Volume | No shift                                                                            |

## Addition of Catalyst



A catalyst affects forward and backward reaction in the same manner.

Equilibrium remains undisturbed but time required to achieve equilibrium reduces.



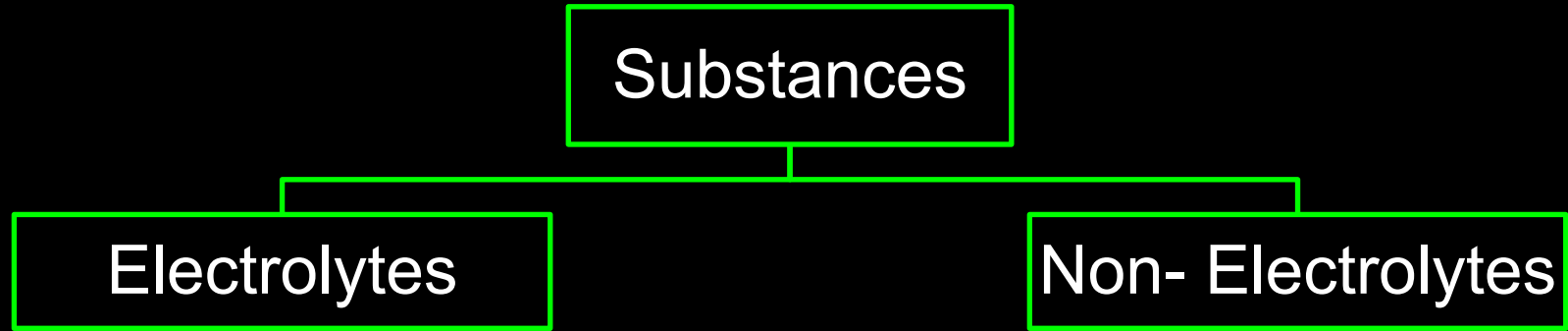
## Summary

- $K_C = \frac{C_C^c \times C_D^d}{C_A^a \times C_B^b}$        $K_P = \frac{p_C^c \times p_D^d}{p_A^a \times p_B^b}$
- $K_P = K_C \times (RT)^{\Delta n_g}$
- If equation is reversed then,  $K' = \frac{1}{K}$
- If equation is multiplied by a number 'n' then,  $K' = K^n$
- If two equations are added  $K' = K_1 \times K_2$

|  |                |
|--|----------------|
|  | Equilibrium    |
|  | Forward shift  |
|  | Backward shift |

- $\Delta_r G^0 = -2.303 \times RT \log K_{eq}$
- Le-Chatelier's Principle: "A change in any of the factors that determine the equilibrium conditions of a system will cause the system to change in such a manner so as to reduce or to counteract the effect of the change".

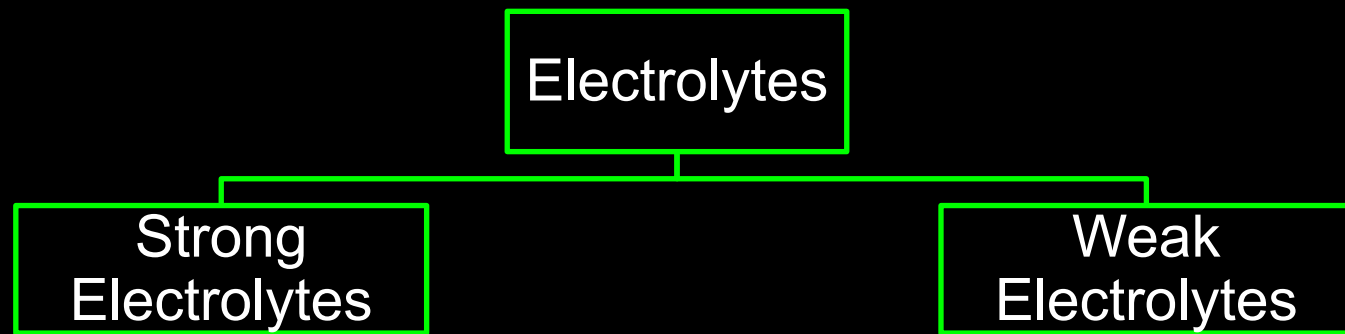
# Types of Substances



- Its aqueous solution conducts electricity.
- Dissociates in corresponding ions on dissolving in water.
- Examples: NaOH, HCl, NaCl etc.
- $\text{NaOH} + x\text{H}_2\text{O} \rightarrow \text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq})$

- Its aqueous solution does not conduct electricity.
- Do not dissociate on dissolving in water.
- Examples: Sugar, urea etc.
- $\text{NH}_2\text{CONH}_2 + x\text{H}_2\text{O} \rightarrow \text{NH}_2\text{CONH}_2(\text{aq})$

# Types of Electrolytes

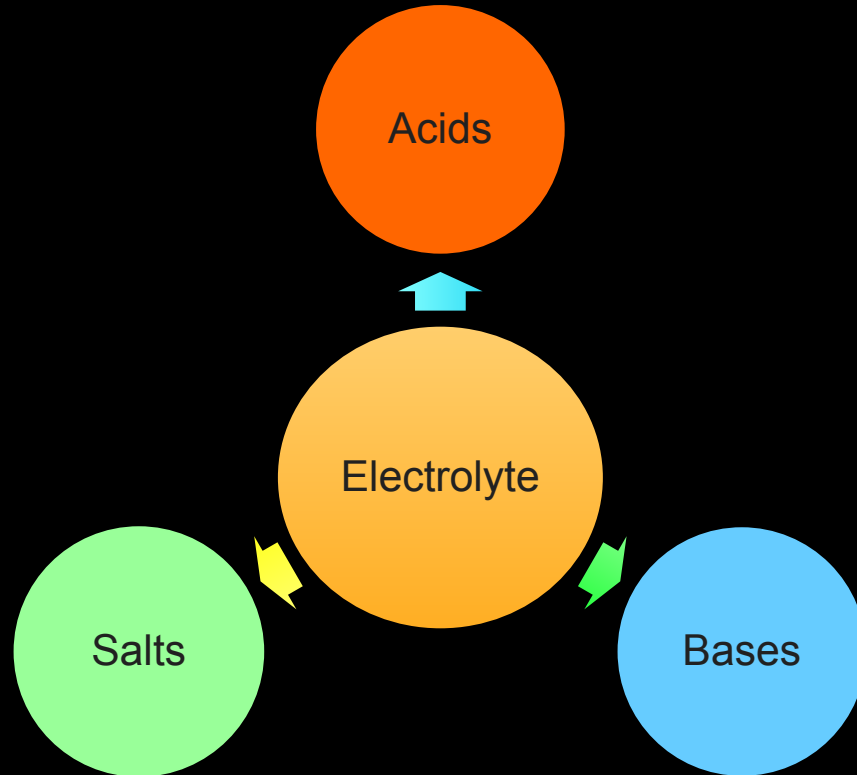


- Dissociates completely on dissolving in water.
- Degree of dissociation  $\alpha \approx 1$ .
- Examples: NaOH, HCl, NaCl etc.
- Dissociation almost reaches to completion hence **no equilibrium** exists.
- $\text{NaOH} + x\text{H}_2\text{O} \rightarrow \text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq})$

- Dissociates partially on dissolving in water.
- Degree of dissociation  $\alpha < 1$ .
- Examples:  $\text{NH}_4\text{OH}$ ,  $\text{C}_2\text{H}_5\text{COOH}$  etc.
- Partial dissociation of substance occurs hence there will be an **ionic equilibrium**.
- $\text{NH}_4\text{OH} + x\text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$

# Electrolytes

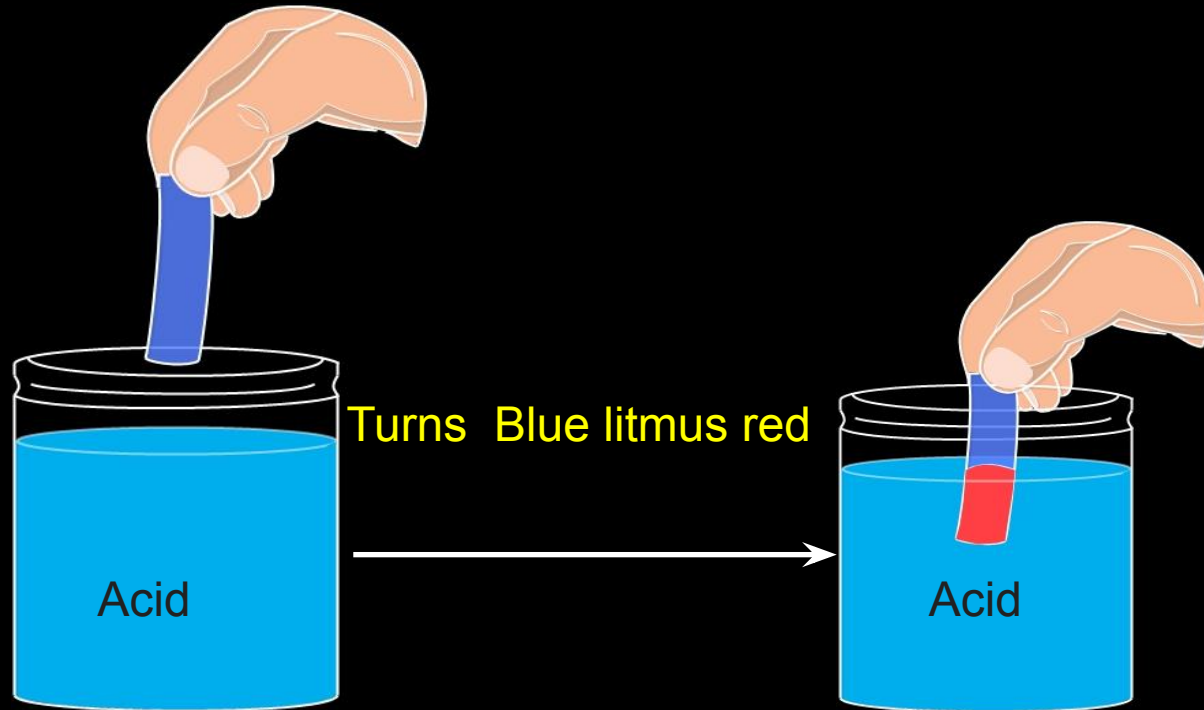
An electrolyte is a chemical compound that dissociates into ions and hence is capable of conducting **electricity**.



# Acids

Derived from a Latin word “acidus” meaning sour

Acid + Metal  $\longrightarrow$  salt + Hydrogen gas



# Acids



Gastric juice (HCl)



Lemon (Citric acid)



Vinegar (Acetic acid)



Tamarind (Tartaric acid)

# Acids



Perchloric acid



Nitric acid



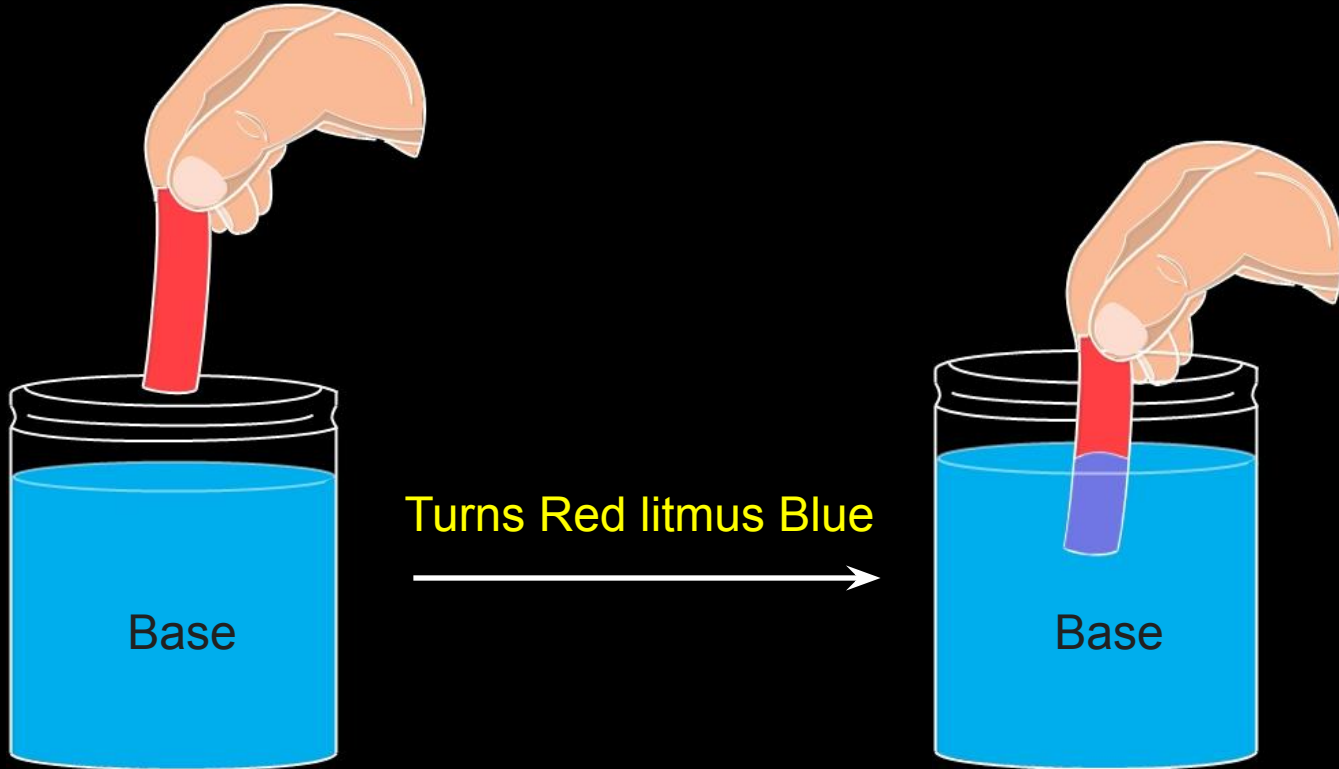
Sulphuric acid



Hydrobromic acid

# Bases

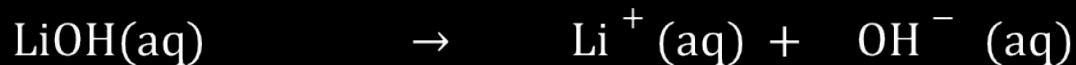
It tastes bitter and feels soapy



# Bases



Sodium Hydroxide



Lithium Hydroxide



Barium Hydroxide

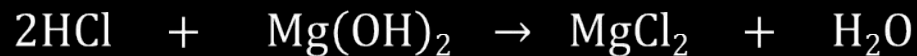


Potassium Hydroxide

# Salts

Reaction of acid and base produces salt and water

Acid + Base  $\rightarrow$  Salt + Water



Sodium Chloride

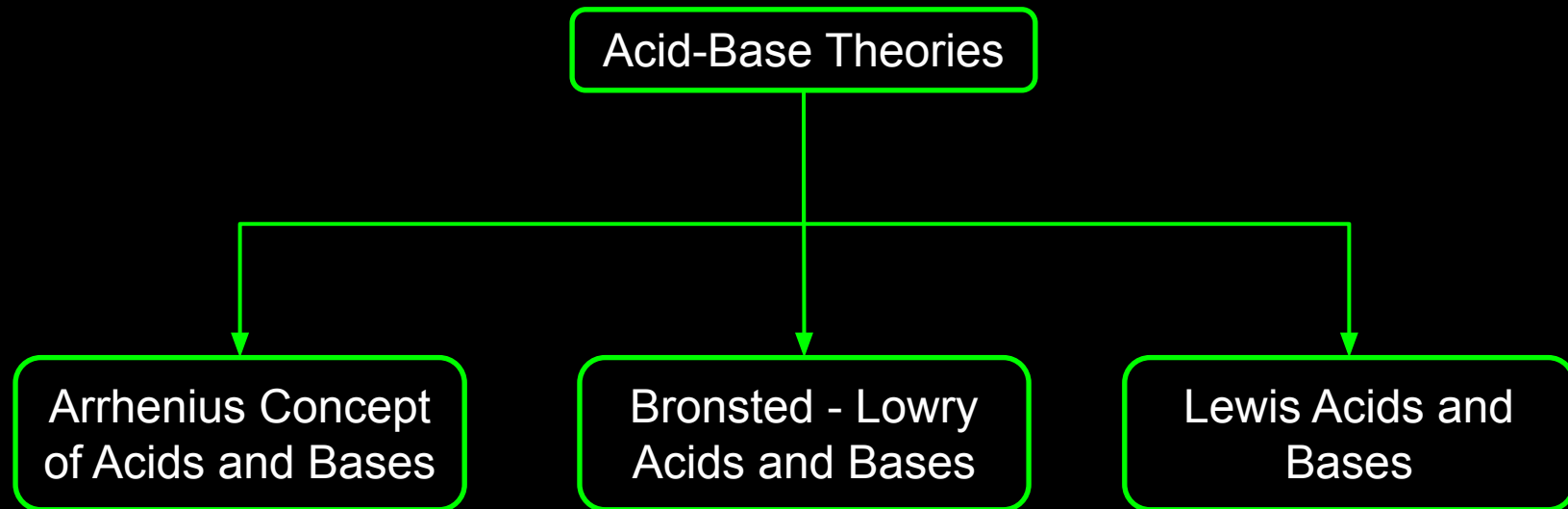


Barium Sulphate



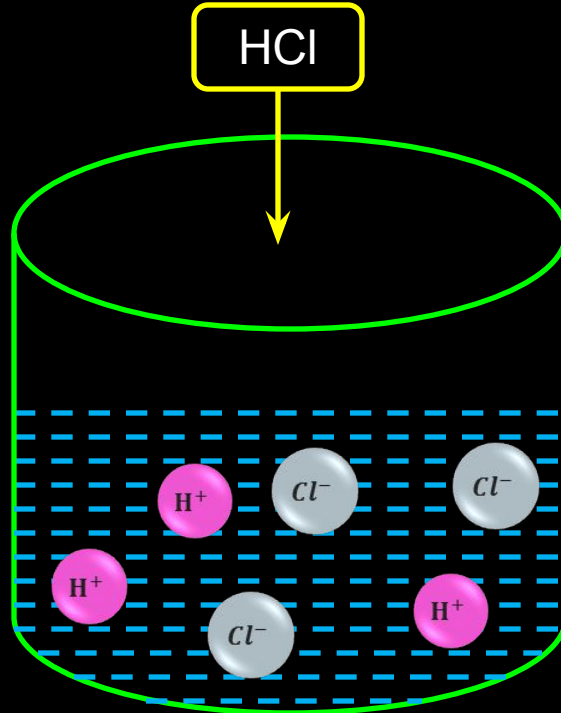
Sodium Nitrate

# Acid – Base Theories

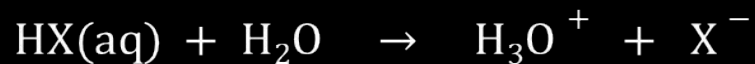
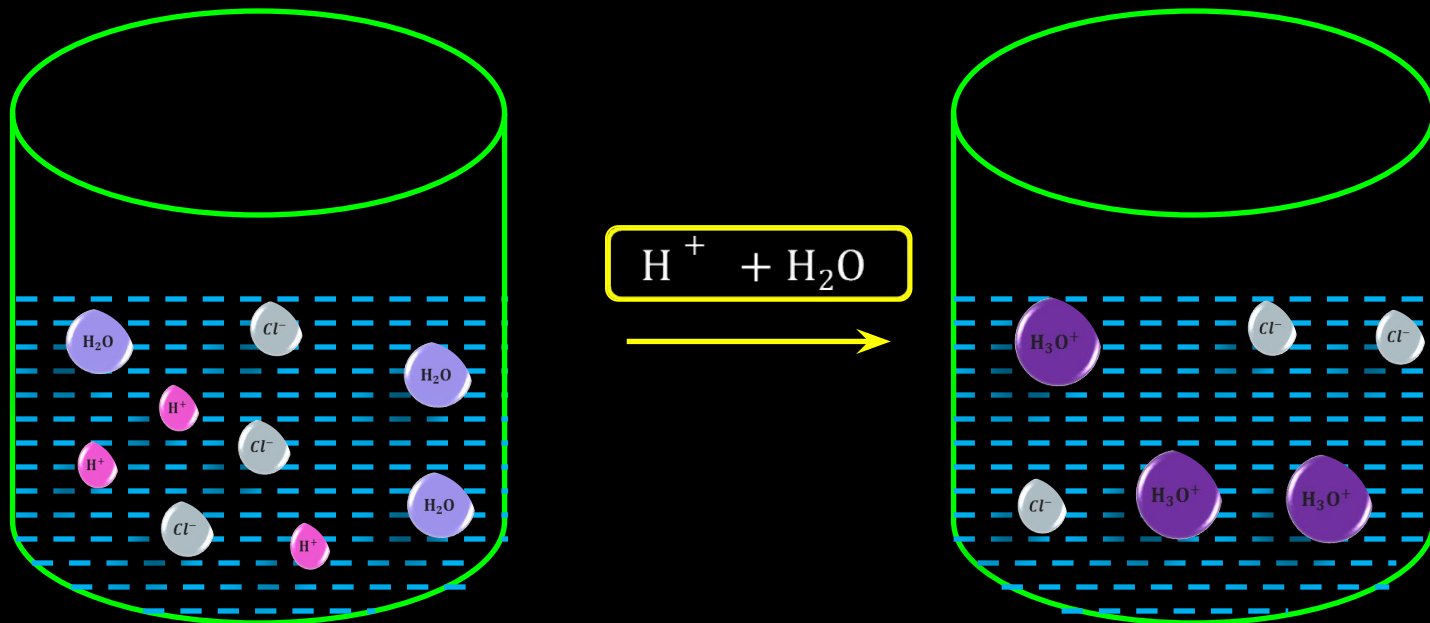
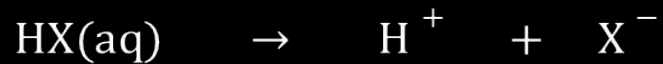


# Arrhenius Concept of Acids and Bases

Acids Dissociates in water to give hydrogen ions (  $\text{H}^+$  )

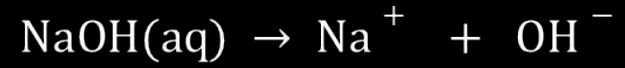
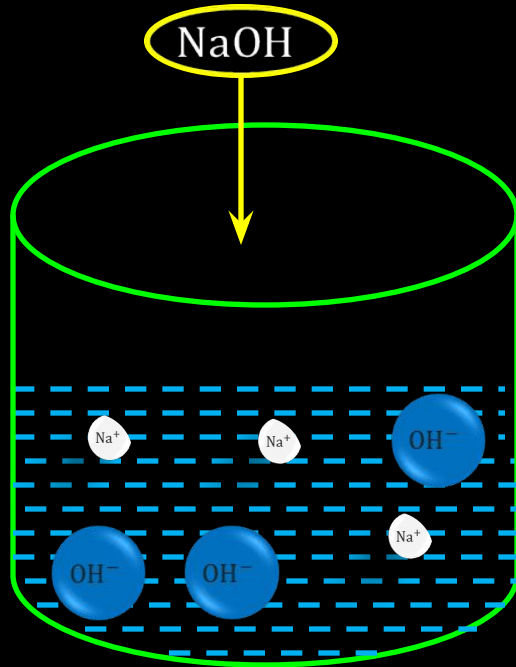


# Arrhenius Concept of Acids and Bases



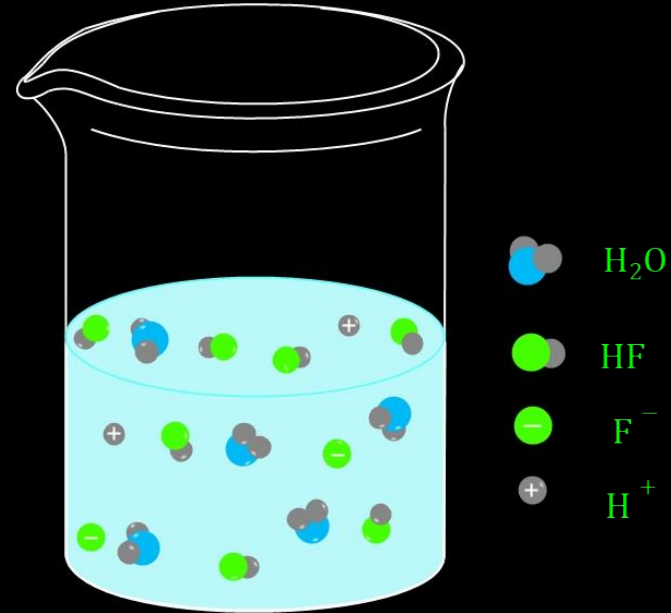
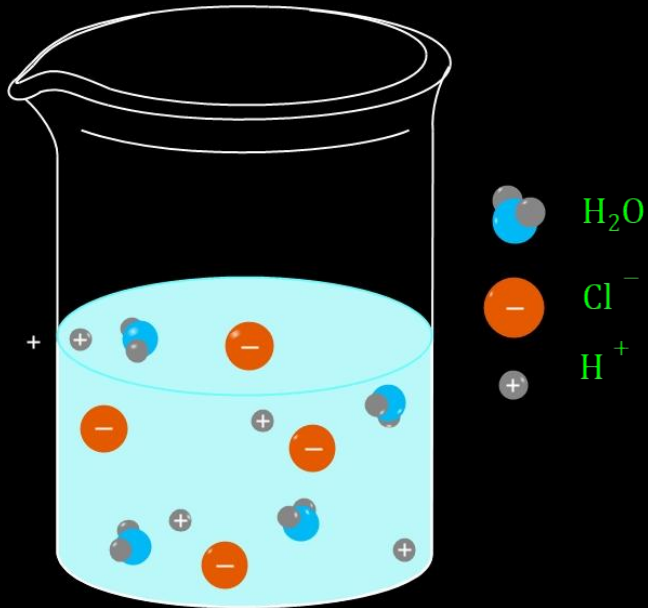
# Arrhenius Concept of Acids and Bases

Bases Dissociates in water to give hydroxyl ions ( $\text{OH}^-$ )



# Limitations of Arrhenius Theory

1. It is limited to aqueous solutions only



## Limitations of Arrhenius Theory

2. Some substance exhibit basic property even though they do not have hydroxide



# Bronsted – Lowry Acids and Bases



Johannes Bronsted



Thomas M. Lowry

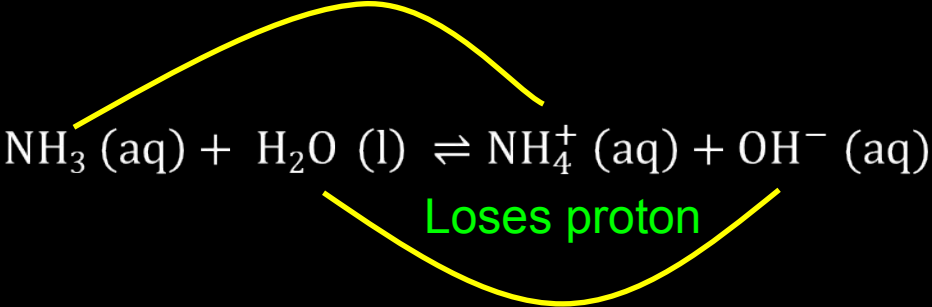
Acid is a substance that is capable of donating a hydrogen ion  $\text{H}^+$  and base is a substance capable of accepting a hydrogen ion,  $\text{H}^+$

OR

Acids are proton donors and bases are proton acceptors

# Bronsted – Lowry Acids and Bases

Adds proton

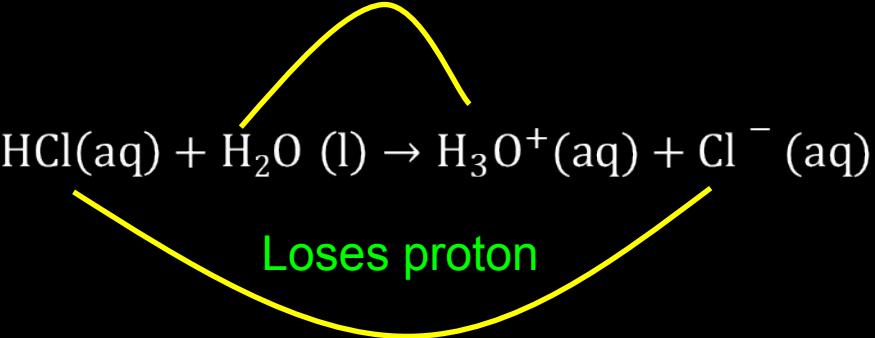


Base                  Acid                  Conjugate acid                  Conjugate base

H<sub>2</sub>O : Bronsted Lowry Acid

NH<sub>3</sub> : Bronsted Lowry Base

Adds proton



Acid                  Base                  Conjugate acid                  Conjugate base

HCl : Bronsted Lowry Acid

H<sub>2</sub>O : Bronsted Lowry Base

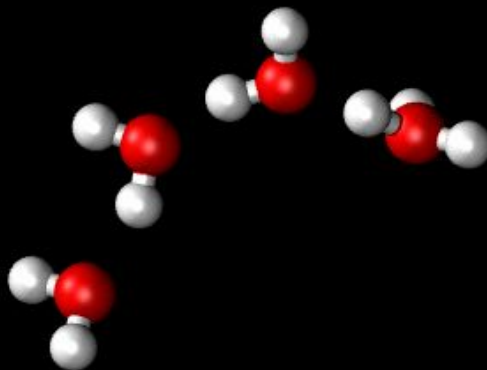
H<sub>2</sub>O : Amphoteric Species, acts as both acid and bases

# Conjugate Pairs

Conjugate acid – base pair



Conjugate acid – base pair



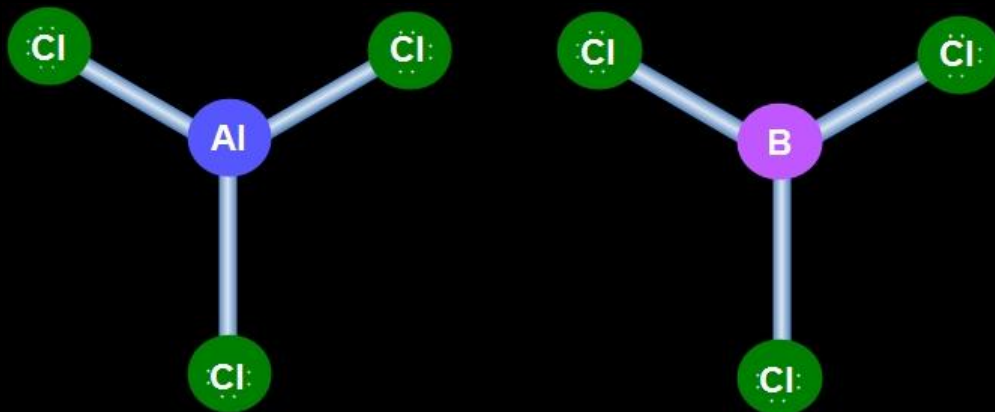
Conjugate acid – base pairs differ by only one  $\text{H}^+$  ion

# Limitations of Bronsted - Lowry Theory

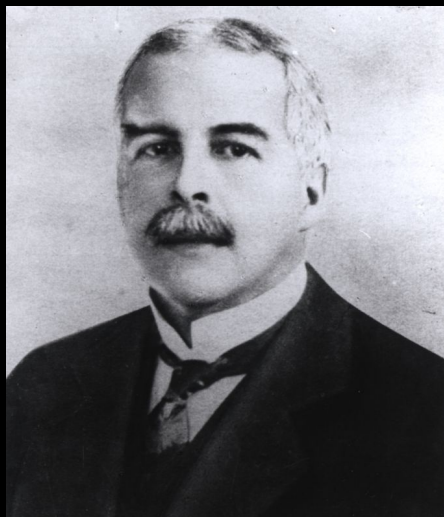
1. It Fails to explain some reactions that do not involve a proton transfer



2. It fails to explain the acidic behavior of electron deficient molecules like  $\text{AlCl}_3$  and  $\text{BCl}_3$

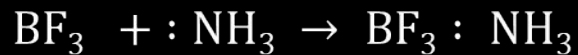


# Lewis Acids and Bases



G. N. Lewis

Acids accept electron pair and bases donate electron pair



| Lewis Acids | Lewis Bases |
|-------------|-------------|
|             |             |
|             |             |
|             |             |

# Concept Test

Ready for Challenge

Q. Classify the following species into Lewis acids and Lewis bases and show how these act as such:

(a)  $\text{OH}^-$  (b)  $\text{F}^-$  (c)  $\text{H}^+$  (d)  $\text{BCl}_3$

Pause the video

Time Duration: 01 min

**Q.** Classify the following species into Lewis acids and Lewis bases and show how these act as such:

(a)  $\text{OH}^-$  (b)  $\text{F}^-$  (c)  $\text{H}^+$  (d)  $\text{BCl}_3$

**Sol.**

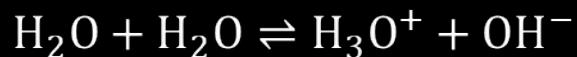
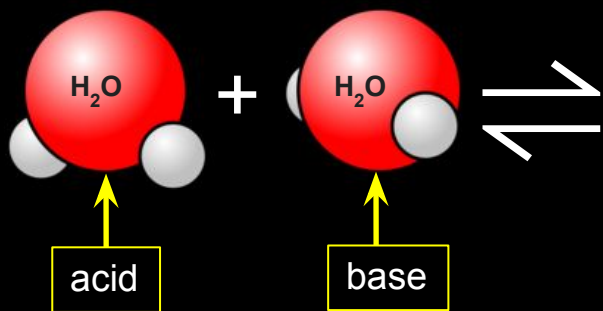
|  |            |                                                                              |
|--|------------|------------------------------------------------------------------------------|
|  | Lewis Base | Donates an electron pair                                                     |
|  | Lewis Base | Donates any of its four electron lone pairs                                  |
|  | Lewis Acid | Accept a lone pair of electron from bases like fluoride ion and hydroxyl ion |
|  | Lewis Acid | Accept lone pair of electron from species like ammonia or amine molecule     |

11C07.3

CV 3

Self Ionization of Water

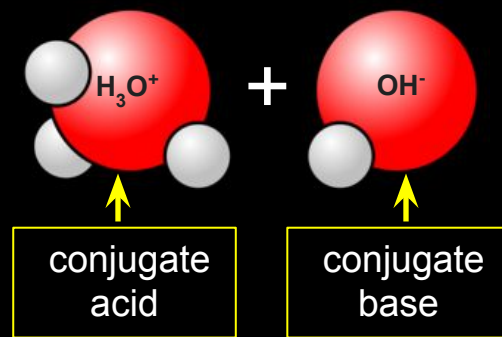
# Self Ionization of Water



$$K = \frac{[\text{H}_3\text{O}^+][\text{OH}^-]}{[\text{H}_2\text{O}]}$$

$K_w = K \times [\text{H}_2\text{O}] = [\text{H}_3\text{O}^+][\text{OH}^-]$   
 Ionic Product of water  
 at 298 K

$$K_w = 1.0 \times 10^{-14} \text{ mol}^2 \text{L}^{-2}$$



$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-]$$

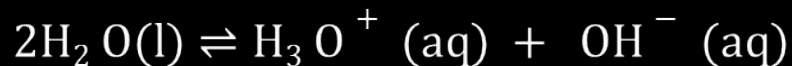
$$[\text{H}_3\text{O}^+] = [\text{OH}^-]$$

$$K_w = [\text{H}_3\text{O}^+]^2 = 10^{-14}$$

$$[\text{H}_3\text{O}^+] = 1.0 \times 10^{-7} \text{ molL}^{-1}$$

**Q.** A solution in water has a hydronium ion concentration of  $2 \times 10^{-6}$  M. What is the concentration of hydroxide ion at  $25^{\circ}\text{C}$ ?

**Sol.** We know that ionic product of water at 298 K is,



$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14}$$

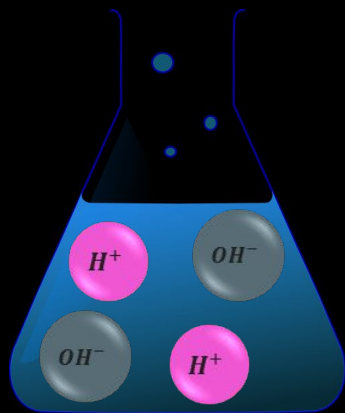
Therefore,

$$[\text{OH}^-] = \frac{K_w}{[\text{H}_3\text{O}^+]}$$

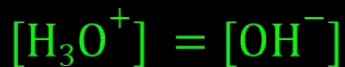
$$= \frac{10^{-14}}{2.0 \times 10^{-6}}$$

$$= 5.0 \times 10^{-9} \text{ molL}^{-1}$$

# Self Ionization of Water



Neutral

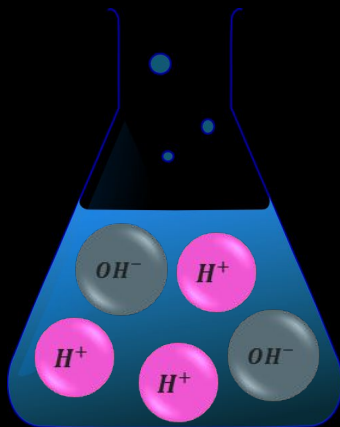


$$[\text{H}_3\text{O}^+] = 10^{-7}$$

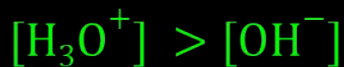
$$\text{pH} = -\log[\text{H}_3\text{O}^+]$$

$$\text{pH} = -\log 10^{-7}$$

$$\text{pH} = 7$$



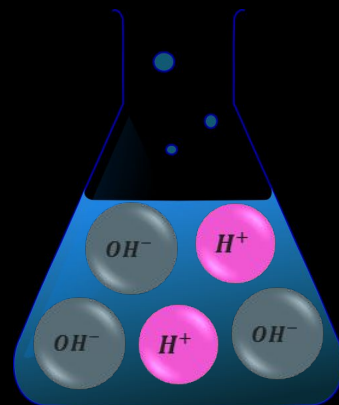
Acidic



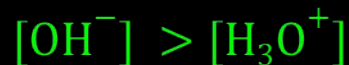
$$[\text{H}_3\text{O}^+] > 10^{-7}$$

$$\text{pH} = -\log[\text{H}_3\text{O}^+]$$

$$\text{pH} < 7$$



Basic



$$[\text{H}_3\text{O}^+] < 10^{-7}$$

$$\text{pH} = -\log[\text{H}_3\text{O}^+]$$

$$\text{pH} > 7$$

# Self Ionization of Water

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-]$$

taking -ve log of both sides

$$-\log K_w = -\log[\text{H}_3\text{O}^+][\text{OH}^-]$$

$$\text{p}K_w = -\log[\text{H}_3\text{O}^+] - \log[\text{OH}^-]$$

$$\text{p}K_w = \text{pH} + \text{pOH}$$

at 298 K

$$K_w = 10^{-14} \Rightarrow \text{p}K_w = 14$$

So,

$$\text{pH} + \text{pOH} = 14$$

$K_w$  depends on temperature, so above equation is applicable only at 298 K.

11C07.3

PSV 1

**Q.** The pH of a sample of vinegar is 3.76. Calculate the concentration of hydrogen ion in it.

**Sol.** Given,

$$\text{pH} = 3.76$$

We know that,

$$\text{pH} = -\log[\text{H}_3\text{O}^+]$$

$$\Rightarrow [\text{H}_3\text{O}^+] = \text{antilog}(-\text{pH})$$

$$[\text{H}_3\text{O}^+] = \text{antilog}(-3.76)$$

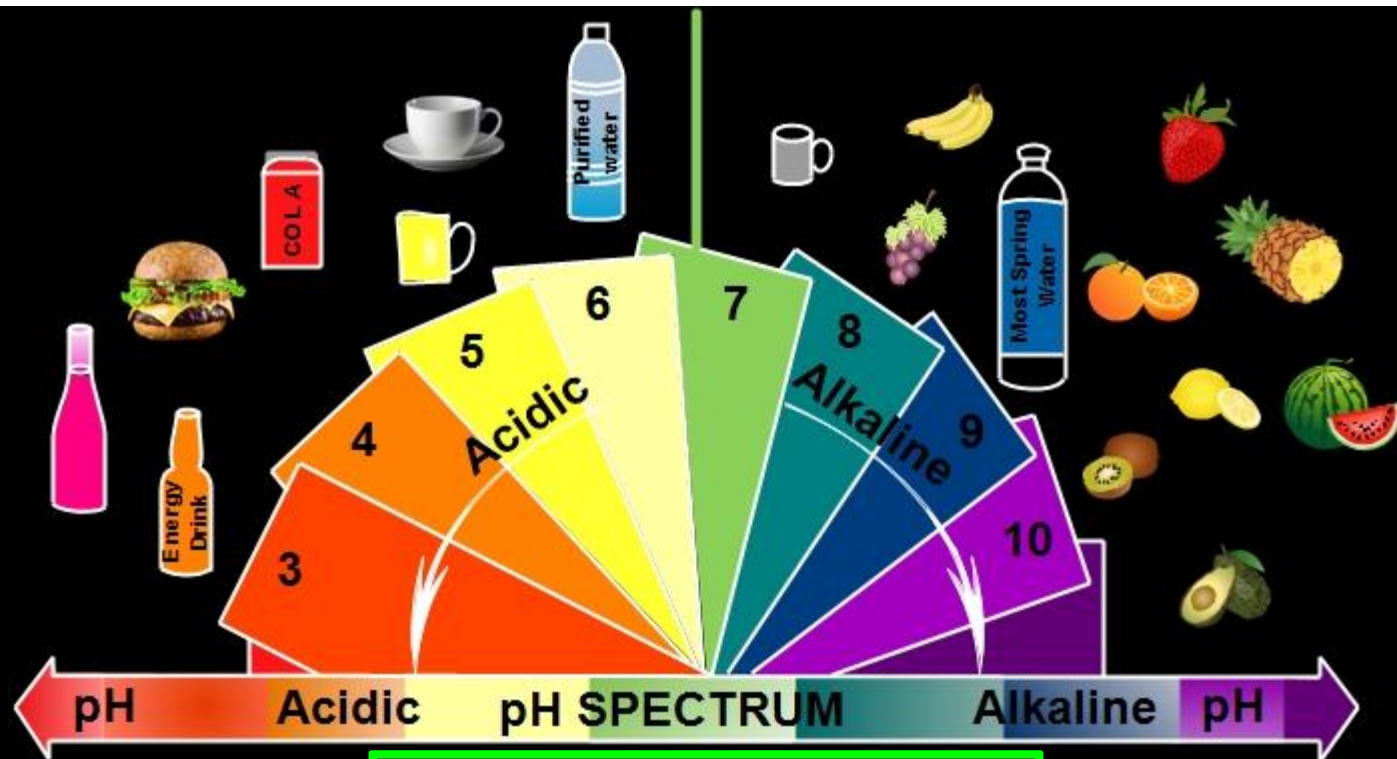
$$= 1.74 \times 10^{-4} \text{ M}$$

11C07.3

CV 4

pH of Strong Acids and Bases

# The pH Scale



Range of pH is from 0 to  $pK_w$

$\frac{pK_w}{2}$  is the neutral point

# The pH Scale

pH – Meter



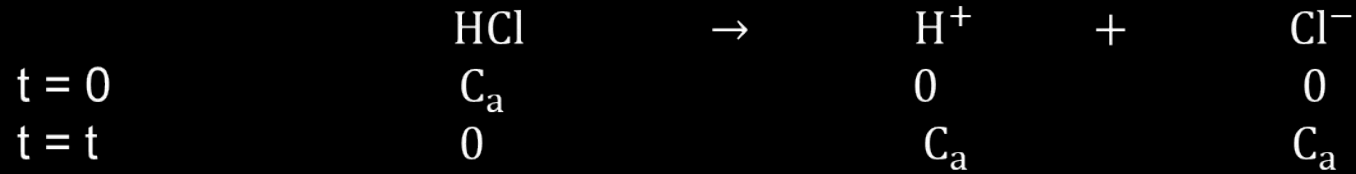
- It is a device that measures pH dependent electrical potential of the solution
- It is used for greater accuracy
- Measures with 0.001 precision

pH – Paper



- It shows different colour in solution with different pH
- pH of a solution can be found roughly with the help of pH paper
- Measures with 0.5 accuracy

# pH of Strong acids and bases



If  $C_a \gg 10^{-7}$  then,

H<sup>+</sup> from dissociation of H<sub>2</sub>O can be neglected

So,  $[H^+] = C_a$

$$pH = -\log C_a$$

Similarly,

In case of strong base

$[OH^-] = C_b$

$$pOH = -\log C_b$$

Q. The concentration of hydrogen ion in a sample of soft drink is  $3.8 \times 10^{-3}$ .  
What is its pH?

Pause the video

Time Duration: 01 min

**Q.** The concentration of hydrogen ion in a sample of soft drink is  $3.8 \times 10^{-3}$ .  
What is its pH?

**Sol.** We know,

$$\begin{aligned}\text{pH} &= -\log[\text{H}^+] \\ &= -\log[3.8 \times 10^{-3}] \\ &= -\{\log[3.8] + \log[10^{-3}]\} \\ &= -\{(0.58) + (-3.0)\} \\ &= 2.42\end{aligned}$$

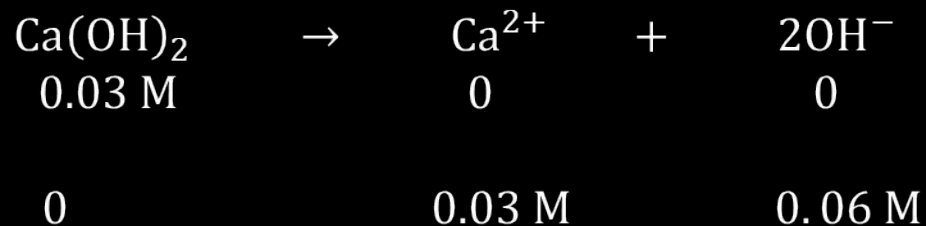
Hence, the soft drink is acidic with a pH of 2.42

11C07.3

PSV 2

Q. Find the pH of 0.03 M  $\text{Ca}(\text{OH})_2$  solution at 298 K.

Sol.



$$\text{pOH} = -\log[\text{OH}^-]$$

$$\text{pOH} = -\log 0.06$$

$$\text{pOH} = 2 - 0.777 = 1.223$$

At 298 K,

$$\text{pH} = 14 - \text{pOH}$$

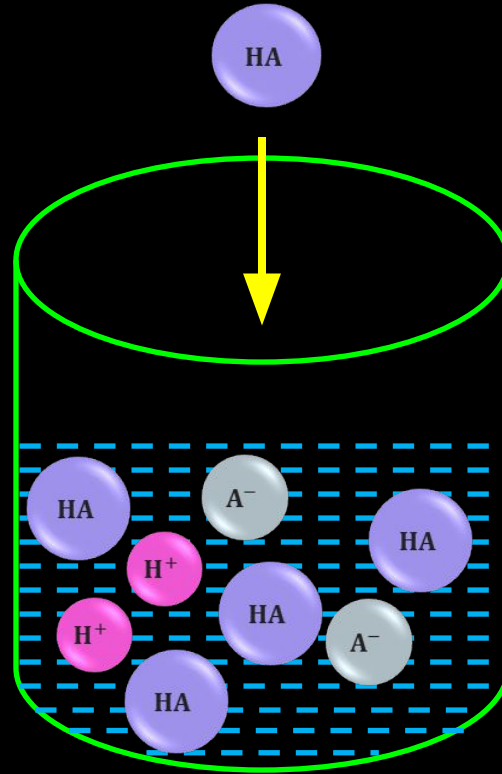
$$\text{pH} = 14 - 1.223 = 12.777$$

11C07.3

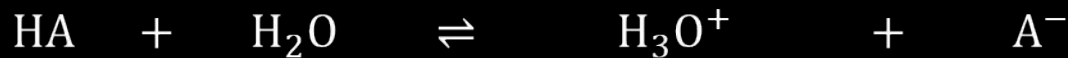
CV 5

pH of weak Acids and Bases

# Dissociation of Weak Acid



## Dissociation of Weak Acid



$$t = 0 \qquad \qquad \text{C} \qquad \qquad \qquad 0 \qquad \qquad \qquad 0$$

Let  $\alpha$  is degree of dissociation of acid at equilibrium

$$t = t_{\text{eq}} \qquad \qquad \text{C}(1 - \alpha) \qquad \qquad \qquad \text{C}\alpha \qquad \qquad \qquad \text{C}\alpha$$

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]} = \frac{\text{C}\alpha \times \text{C}\alpha}{\text{C}(1 - \alpha)}$$

$K_a$  is dissociation constant of acid

Since  $\alpha \ll 1$

$$K_a = \text{C}\alpha^2$$

## Dissociation constant of Weak Acid

$$K_a = C\alpha^2$$

$$[H_3O^+] = C\alpha = \sqrt{K_a \times C}$$

Taking -ve log of both sides

$$-\log[H_3O^+] = -\log\sqrt{K_a \times C}$$

$$pH = \frac{1}{2}(-\log K_a - \log C)$$

$$pH = \frac{1}{2}(pK_a - \log C)$$

**Note:**  $K_a$  is a measure of strength of acid HA i.e. larger the value of  $K_a$ , the stronger is the acid

**Q.** Calculate the pH of 0.01M  $\text{CH}_3\text{COOH}$  solution. Also find its degree of dissociation.  
( $K_a$  of  $\text{CH}_3\text{COOH}$  is  $1.8 \times 10^{-5}$ )

**Sol.** Given:  $K_a$  of  $\text{CH}_3\text{COOH}$  is  $1.8 \times 10^{-5}$

$$\text{p}K_a = -\log K_a$$

$$\text{p}K_a = -\log 1.8 \times 10^{-5}$$

$$\text{p}K_a = 4.74$$

$$\text{pH} = \frac{1}{2} (\text{p}K_a - \log C)$$

$$\text{pH} = \frac{1}{2} (4.74 - \log(0.01))$$

$$\text{pH} = \frac{1}{2} (4.74 + 2) = \frac{1}{2} \times 6.74$$

$$\text{pH} = 3.37$$

$$K_a = C\alpha^2$$

$$\alpha = \sqrt{\frac{K_a}{C}}$$

$$\alpha = \sqrt{\frac{1.8 \times 10^{-5}}{0.01}}$$

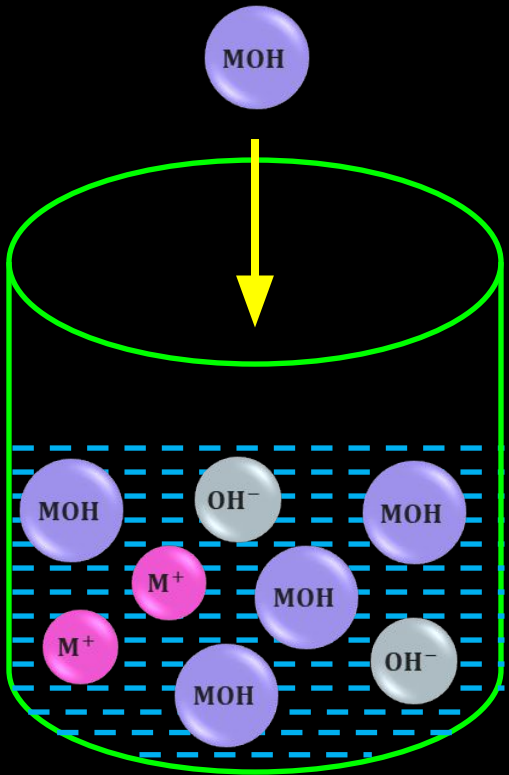
$$\alpha = \sqrt{1.8 \times 10^{-3}}$$

$$\alpha = 4.24 \times 10^{-2}$$

# Ionization constant of Weak Acid

| Acid              | Formula |  |  |
|-------------------|---------|--|--|
| Hydrofluoric Acid |         |  |  |
| Nitrous Acid      |         |  |  |
| Formic Acid       |         |  |  |
| Niacin            |         |  |  |
| Acetic acid       |         |  |  |
| Benzoic acid      |         |  |  |
| Hypochlorous acid |         |  |  |
| Hydrocyanic acid  |         |  |  |
| Phenol            |         |  |  |

# Dissociation of Weak Base



## Dissociation of Weak Base



$$t = 0 \qquad \qquad \qquad C \qquad \qquad 0 \qquad \qquad 0$$

Let  $\alpha$  be degree of dissociation of base at equilibrium

$$t = t_{\text{eq}} \qquad \qquad C(1 - \alpha) \qquad \qquad C\alpha \qquad \qquad C\alpha$$

$$K_b = \frac{[\text{OH}^-][\text{M}^+]}{[\text{MOH}]} = \frac{C\alpha \times C\alpha}{C(1 - \alpha)}$$

$K_b$  is dissociation constant of base

Since  $\alpha \ll 1$

$$K_b = C\alpha^2$$

## Dissociation constant of Weak Base

$$K_b = C\alpha^2$$

$$[\text{OH}^-] = C\alpha = \sqrt{K_b \times C}$$

Taking -ve log of both sides

$$-\log[\text{OH}^-] = -\log\sqrt{K_b \times C}$$

$$\text{pOH} = \frac{1}{2}(-\log K_b - \log C)$$

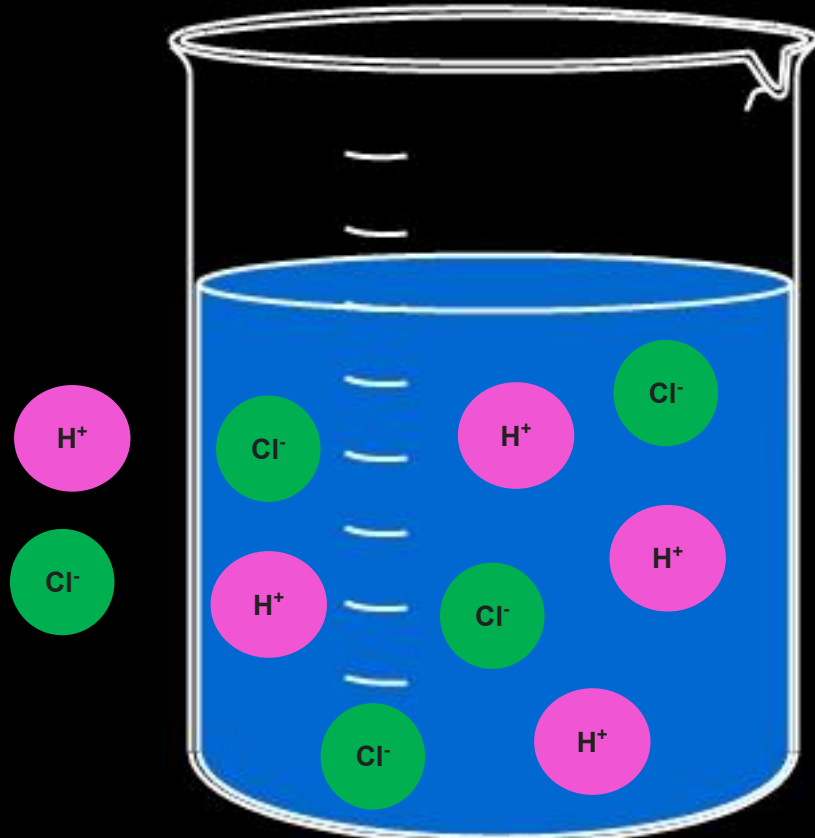
$$\text{pOH} = \frac{1}{2}(\text{p}K_b - \log C)$$

**Note:**  $K_b$  is a measure of strength of base MOH i.e. larger the value of  $K_b$ , the stronger is the base

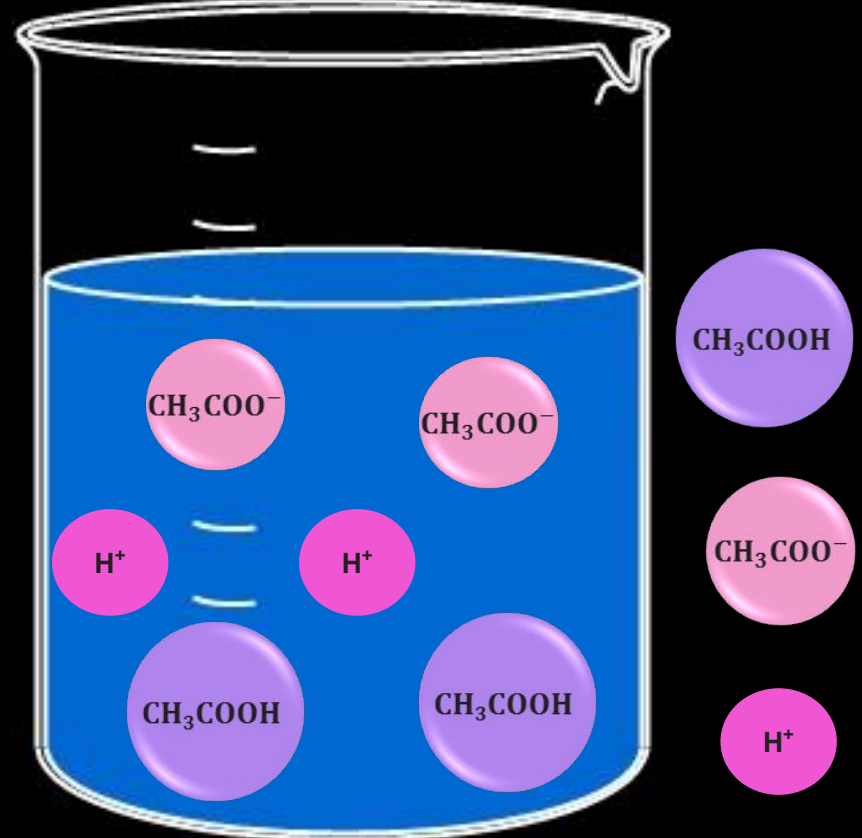
# Ionization of Weak Base

| Base          | Formula              | Ionization constant ,<br>$K_b$ | $pK_b$ |
|---------------|----------------------|--------------------------------|--------|
| Dimethylamine | $(CH_3)_2NH$         | $5.4 \times 10^{-4}$           | 3.26   |
| Triethylamine | $(C_2H_5)_3N$        | $6.45 \times 10^{-5}$          | 4.19   |
| Ammonia       | $NH_3$ or $NH_4OH$   | $1.77 \times 10^{-5}$          | 4.75   |
| Quinine       | $C_{20}H_{24}N_2O_2$ | $1.10 \times 10^{-6}$          | 5.95   |
| Pyridine      | $C_5H_5N$            | $1.77 \times 10^{-9}$          | 8.752  |
| Aniline       | $C_6H_5NH_2$         | $4.27 \times 10^{-10}$         | 9.36   |
| Urea          | $CO(NH_2)_2$         | $1.3 \times 10^{-14}$          | 13.88  |

# Factors affecting acidic strength



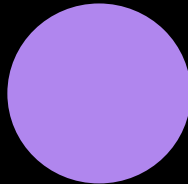
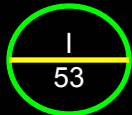
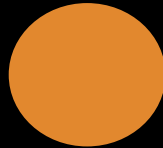
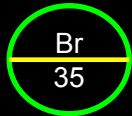
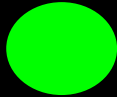
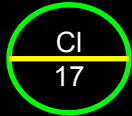
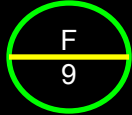
$HCl$   
(Dissociates completely)



$CH_3COOH$   
(Does not Dissociate completely)

# Factors affecting acidic strength

Group 17



Size Increases



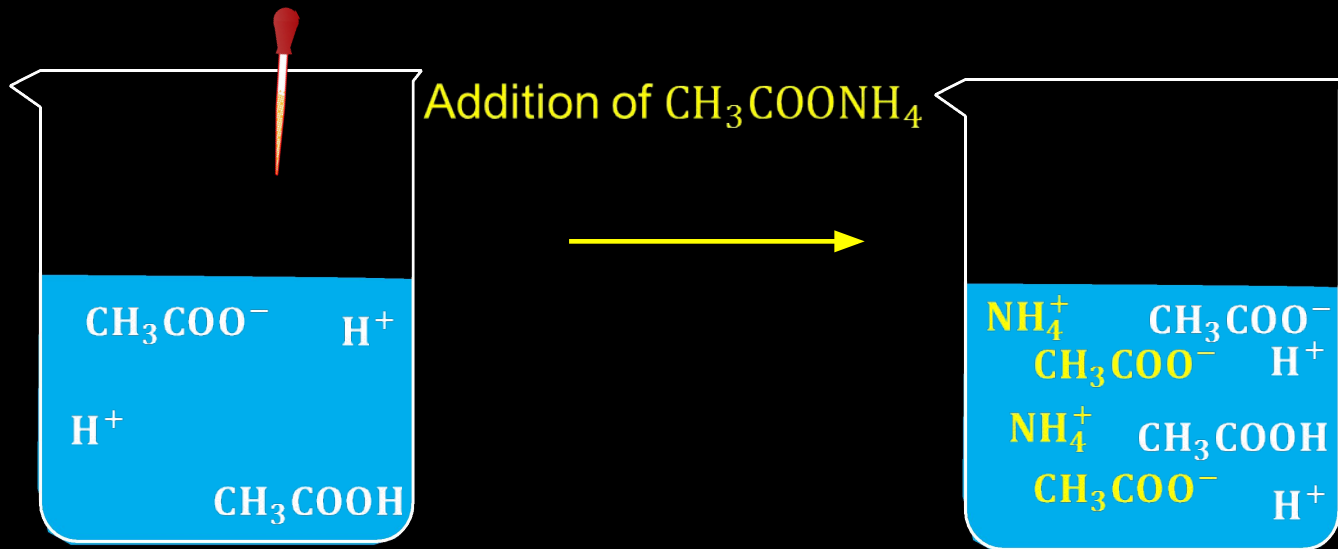
11C07.3

CV 6

Common Ion Effect

# Common Ion effect

According to Le Chateliers principle



On addition of common ion in solution of weak electrolyte its dissociation decreases

# Common Ion effect

|                                                                                          |                                   |   |                     |   |                                   |
|------------------------------------------------------------------------------------------|-----------------------------------|---|---------------------|---|-----------------------------------|
|                                                                                          | HA(aq)                            | ⇌ | H <sup>+</sup> (aq) | + | A <sup>-</sup> (aq)               |
| Initial conc.                                                                            | C <sub>a</sub>                    |   | 0                   |   | C <sub>s</sub>                    |
| Change in conc.                                                                          | -C <sub>i</sub>                   |   | +C <sub>i</sub>     |   | +C <sub>i</sub>                   |
| Equilibrium conc.                                                                        | C <sub>a</sub> - C <sub>a</sub> α |   | C <sub>a</sub> i    |   | C <sub>s</sub> + C <sub>a</sub> α |
| $K_a = \frac{[H^+][A^-]}{[HA]}$ $= \frac{(C_s + C_a\alpha)(C_a\alpha)}{C_a - C_a\alpha}$ |                                   |   |                     |   |                                   |

## Common Ion effect

$$K_a = \frac{(C_s + C_a \alpha)(C_a \alpha)}{(C_a - C_a \alpha)}$$

$((C_s + C_a \alpha) \approx C_s$  and  $(C_a - C_a \alpha) \approx C_a$  (as  $K_a$  for weak acid is very small,  $\alpha \ll 1$ )

Therefore,

$$K_a = \frac{(C_s + C_a \alpha)(C_a \alpha)}{(C_a - C_a \alpha)}$$

$$= \alpha \times C_s$$

$$\alpha = \frac{K_a}{C_s}$$

**Q.** Find the degree of dissociation of 0.01 M  $\text{CH}_3\text{COOH}$  solution. Also find its degree of dissociation in 0.01 M  $\text{CH}_3\text{COONa}$  solution.  $K_a$  for  $\text{CH}_3\text{COOH}$  is  $1.8 \times 10^{-5}$ .

**Sol.**

For 0.01 M  $\text{CH}_3\text{COOH}$  solution

$$\alpha = \sqrt{\frac{K_a}{C_a}}$$

$$\alpha = \sqrt{\frac{1.8 \times 10^{-5}}{0.01}}$$

$$\alpha = 4.24 \times 10^{-2}$$

In 0.01 M  $\text{CH}_3\text{COONa}$

$$\alpha' = \frac{K_a}{C_s}$$

$$\alpha' = \frac{1.8 \times 10^{-5}}{0.01}$$

$$\alpha' = 1.8 \times 10^{-3}$$

## Summary

- All substances that conduct electricity in aqueous solutions are called electrolytes. Acids, bases and salts are electrolytes as they produce cations and anions on dissociation or ionization.

| Arrhenius Theory                                                               | Bronsted-Lowry Theory                                 | Lewis Acids and Bases                                                  |
|--------------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------------------|
| Acids give hydrogen ion while bases produce hydroxyl ions in aqueous solutions | Acids donate hydrogen while bases are proton acceptor | Acids act as electron pair acceptor while base are electron pair donor |

- The pH scale ( $\text{pH} = -\log[\text{H}^+]$ ) for the hydrogen ion concentration (activity) has been introduced and extended to other quantities ( $\text{pOH} = -\log[\text{OH}^-]$ );
- The ionization of water has been considered and the equation :  $\text{pH} + \text{pOH} = \text{pK}_w$  is always satisfied.

## Reference Questions

NCERT In-text Questions: 12 to 24

NCERT Exercises: 7.35 to 7.54

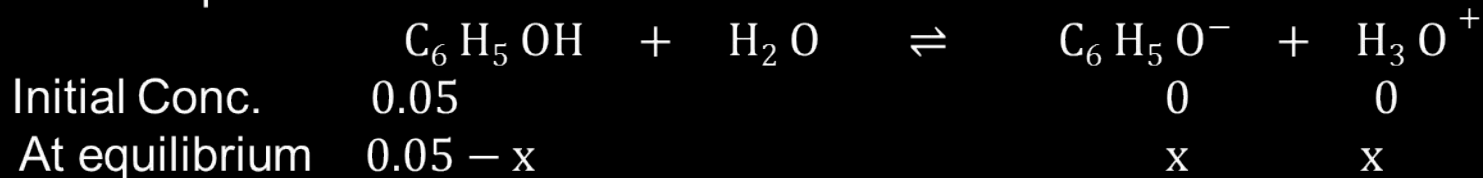
Workbook Question: 8, 9, 10, 11, 13, 14, 15, 16

11C07.3

PSV 3

**Q.** The ionization constant of phenol is  $1.0 \times 10^{-10}$ . What is the concentration of phenolate ion in 0.05 M solution of phenol? Also find its degree of dissociation in 0.01 M phenolate ion solution.

**Sol.** Ionization of phenol:



$$K_a = \frac{[\text{C}_6\text{H}_5\text{O}^-][\text{H}_3\text{O}^+]}{[\text{C}_6\text{H}_5\text{OH}]}$$

$$K_a = \frac{x \times x}{(0.05 - x)}$$

We know,  $x \ll 0.05$

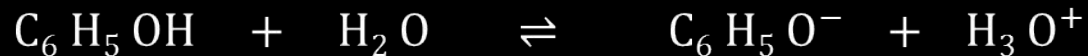
$$K_a \times 0.05 = x^2$$

Therefore,

$$\begin{aligned} x &= \sqrt{1 \times 10^{-10} \times 0.05} \\ &= \sqrt{5 \times 10^{-12}} \end{aligned}$$

$$[\text{C}_6\text{H}_5\text{O}^-] = [\text{H}_3\text{O}^+] = 2.2 \times 10^{-6} \text{ M}$$

Let  $\alpha$  is the degree of ionization of phenol in presence of 0.01 M phenolate ion



Initial Conc.      0.05

0

0

At equilibrium     $0.05 - 0.05\alpha$

$0.01 + 0.05\alpha$

$0.05\alpha$

$$[\text{C}_6\text{H}_5\text{OH}] = 0.05 - 0.05\alpha \approx 0.05\text{M}$$

$$[\text{C}_6\text{H}_5\text{O}^-] = 0.01 + 0.05\alpha \approx 0.01\text{M}$$

$$[\text{H}_3\text{O}^+] = 0.05\alpha$$

$$K_a = \frac{[\text{C}_6\text{H}_5\text{O}^-][\text{H}_3\text{O}^+]}{[\text{C}_6\text{H}_5\text{OH}]}$$

$$K_a = \frac{(0.01)(0.05\alpha)}{0.05}$$

$$1.0 \times 10^{-10} = 0.01\alpha$$

$$\alpha = 1 \times 10^{-8}$$

11C07.4

# Salt Hydrolysis

# 11C07.4 – Salt Hydrolysis

## Learning Objective

Types of salt

Hydrolysis of salt of strong acid and weak base

Hydrolysis of salt of weak acid and strong base

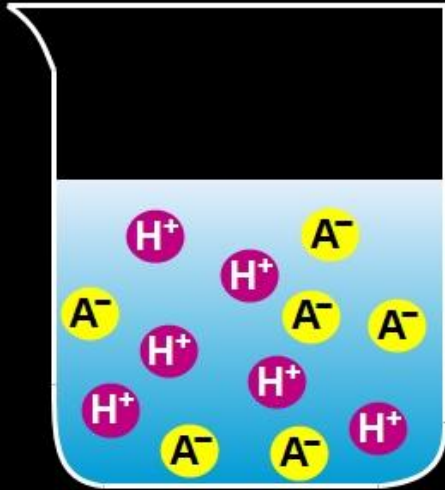
Hydrolysis of salt of weak acid and weak base

11C07.4

CV 1

Types of salt

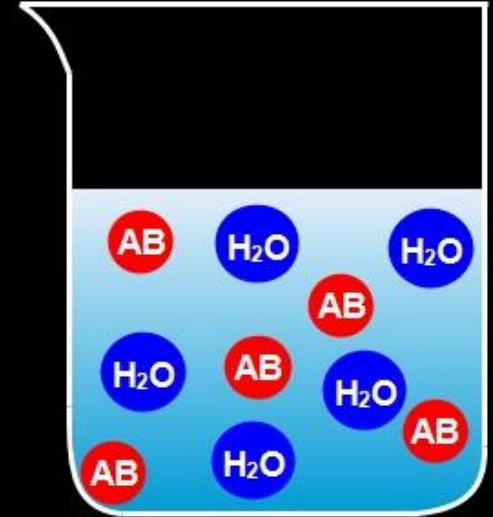
# Types of salt



HA  
(Acid)



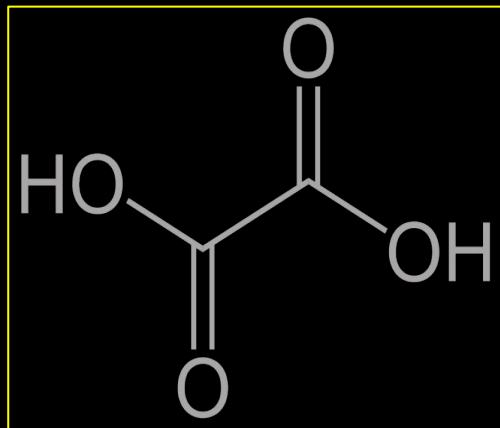
BOH  
(Base)



AB  
(Salt)

## Di- and polybasic acids

They have more than one ionizable proton per molecule of acid

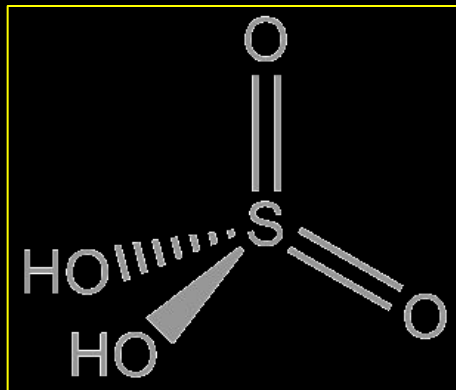


Oxalic acid

$$K_{a_1} = 5.9 \times 10^{-2}$$

$$K_{a_2} = 6.4 \times 10^{-5}$$

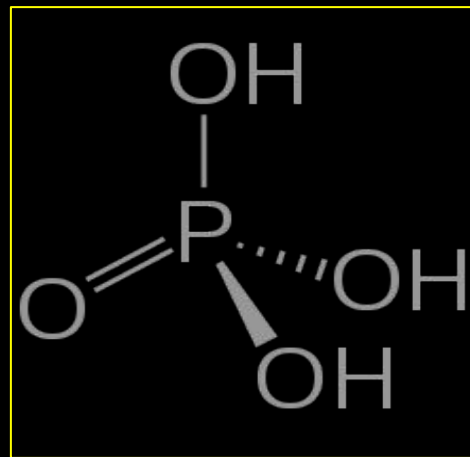
Also known as polyprotic acid



Sulphuric acid

$$K_{a_1} = \text{Very large}$$

$$K_{a_2} = 1.2 \times 10^{-2}$$



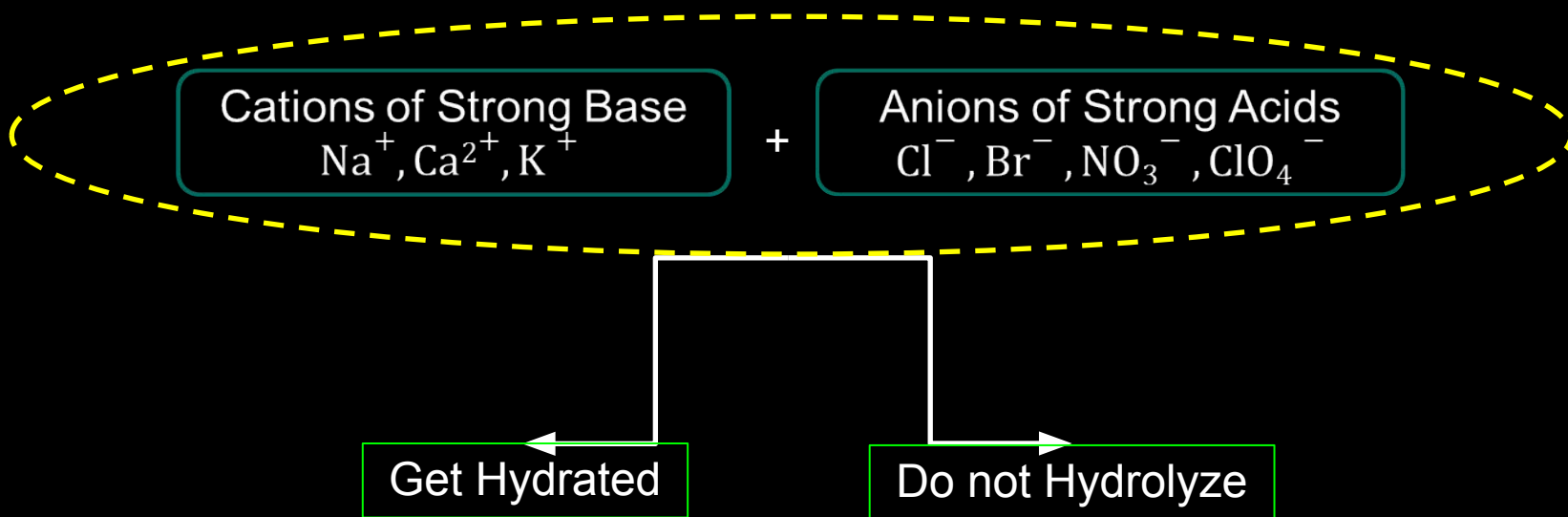
Phosphoric acid

$$K_{a_1} = 7.5 \times 10^{-3}$$

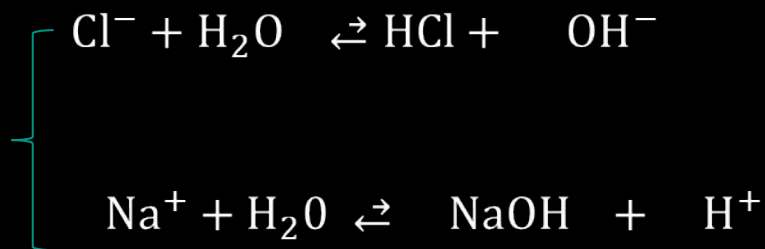
$$K_{a_2} = 6.2 \times 10^{-8}$$

$$K_{a_3} = 4.2 \times 10^{-13}$$

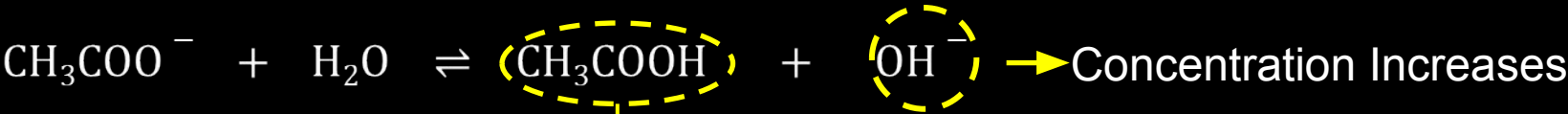
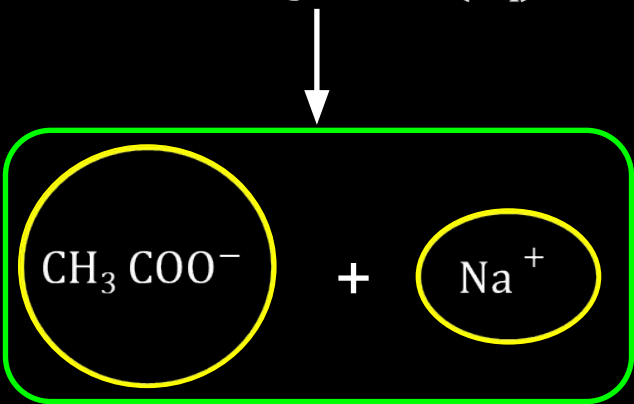
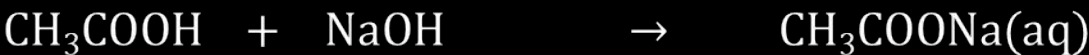
# Salts of Strong acid and Strong Base



Backward Reaction

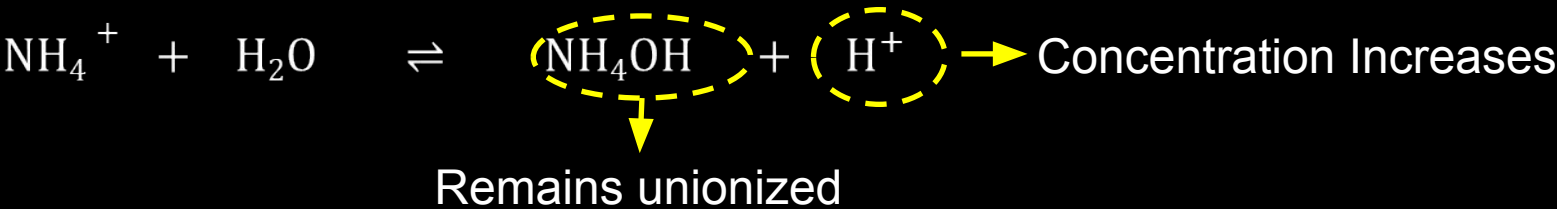
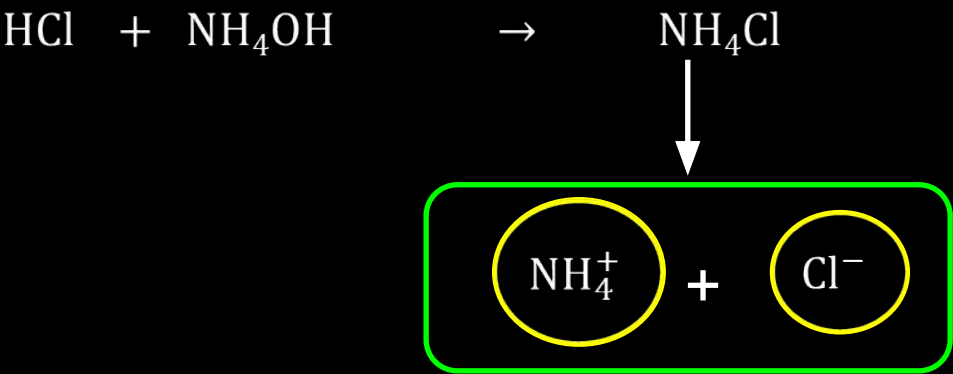


# Salts of Weak acid and Strong Base

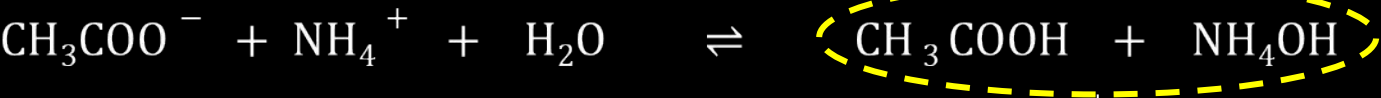
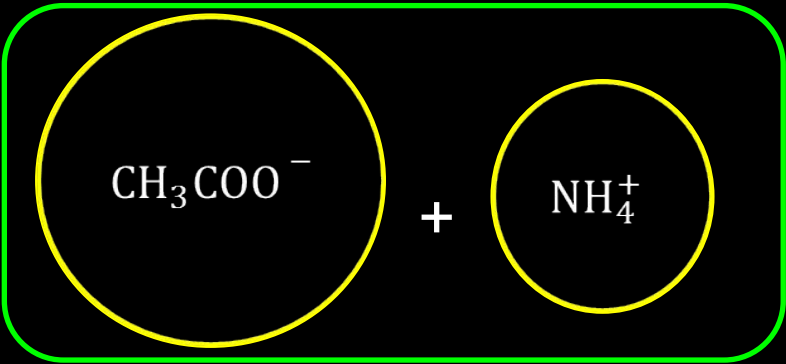
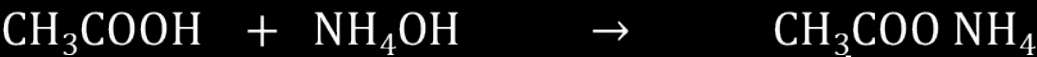


$\downarrow$   
Remains unionized

# Salts of Strong acid and Weak Base



# Salts of Weak acid and Weak Base

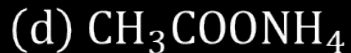
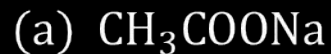


Partial Dissociation

# Concept Test

Ready for challenge

**Q.** Which of the following salts will not hydrolyse?



**Sol.** (c)  $\text{BaCl}_2$

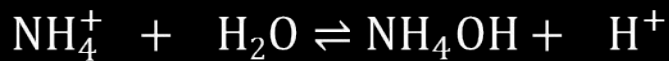
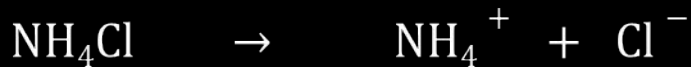
$\text{BaCl}_2$  will not hydrolyse as it is a salt of strong acid  $\text{HCl}$  and strong base  $\text{Ba}(\text{OH})_2$ .

11C07.4

CV 2

Hydrolysis of salt of strong acid and weak base

## Hydrolysis of salt of strong acid and weak base (Cationic hydrolysis)



C(1-h)

Ch

Ch

$$\begin{aligned} K_h &= \frac{[\text{NH}_4\text{OH}][\text{H}^+]}{[\text{NH}_4^+]} \times \frac{[\text{OH}^-]}{[\text{OH}^-]} \\ &= \frac{[\text{NH}_4\text{OH}]}{[\text{NH}_4^+][\text{OH}^-]} \times [\text{H}^+][\text{OH}^-] \end{aligned}$$

$$\left\{ \begin{array}{l} \text{NH}_4\text{OH} \rightleftharpoons \text{NH}_4^+ + \text{OH}^- \\ K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_4\text{OH}]} \end{array} \right\}$$
$$\text{Ch}^2 = K_h = \frac{K_w}{K_b}$$

$$[\text{H}^+] = \sqrt{\text{C}^2 h^2} = \sqrt{\frac{K_w \text{C}}{K_b}}$$

$$\text{pH} = \frac{1}{2} (\text{p}K_w - \text{p}K_b - \log \text{C})$$

at 298 K,

$$\text{pH} = 7 - \frac{1}{2} (\text{p}K_b + \log \text{C})$$

11C07.4

PSV 1

**Q.** The dissociation constant of  $\text{NH}_3$  is  $1.8 \times 10^{-5}$ . If concentration of  $\text{NH}_4\text{Cl}$  is  $10^{-4}\text{M}$  then, what will be its degree of hydrolysis?

**Sol.** We Know that,

$$Ch^2 = K_h = \frac{K_w}{K_b}$$

Therefore,

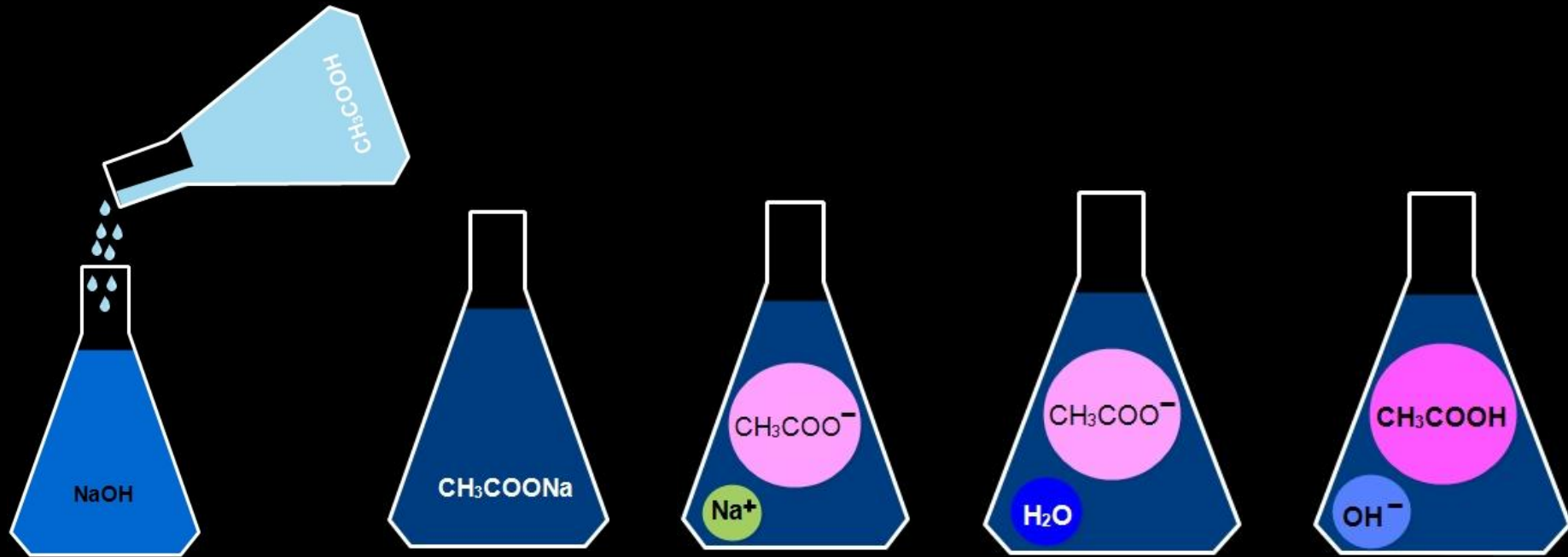
$$\begin{aligned} h &= \sqrt{\frac{K_w}{C \times K_b}} \\ &= \sqrt{\frac{10^{-14}}{1.8 \times 10^{-5} \times 1 \times 10^{-4}}} \\ &= 2.35 \times 10^{-3} \end{aligned}$$

11C07.4

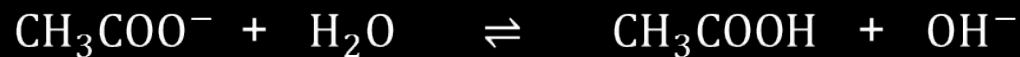
CV 3

Hydrolysis of strong base and weak acid

# Hydrolysis of strong base and weak Acid



## Hydrolysis of strong base and weak Acid



$$C(1 - h)$$

$$Ch$$

$$Ch$$

$$K_h = \frac{[\text{CH}_3\text{COOH}][\text{OH}^-]}{[\text{CH}_3\text{COO}^-]} \times \frac{[\text{H}^+]}{[\text{H}^+]}$$

$$= \frac{[\text{CH}_3\text{COOH}]}{[\text{CH}_3\text{COO}^-][\text{H}^+]} \times [\text{H}^+][\text{OH}^-]$$

$$Ch^2 = K_h = \frac{K_w}{K_a}$$

$$[\text{OH}^-] = ch = \sqrt{\frac{K_w C}{K_a}}$$

$$\text{pOH} = 7 - \frac{1}{2}(\text{p}K_a + \log C)$$

at 298 K,

$$\text{pH} = 14 - \text{pOH}$$

$$\text{pH} = 7 + \frac{1}{2}(\text{p}K_a + \log C)$$

11C07.4

PSV 2

**Q.** A weak acid HX has the dissociation constant value equal to  $1 \times 10^{-5}$  M. It forms a salt NaX on reaction with an alkaline solution. What will be the degree of hydrolysis of 0.1 M solution of NaX?

**Sol.** For a salt of weak acid and a strong base,

$$\begin{aligned}K_h &= \frac{K_w}{K_a} = Ch^2 \\h &= \sqrt{\frac{K_w}{C \times K_a}} \\&= \sqrt{\frac{10^{-14}}{10^{-5} \times 0.1}} \\&= 0.0001\end{aligned}$$

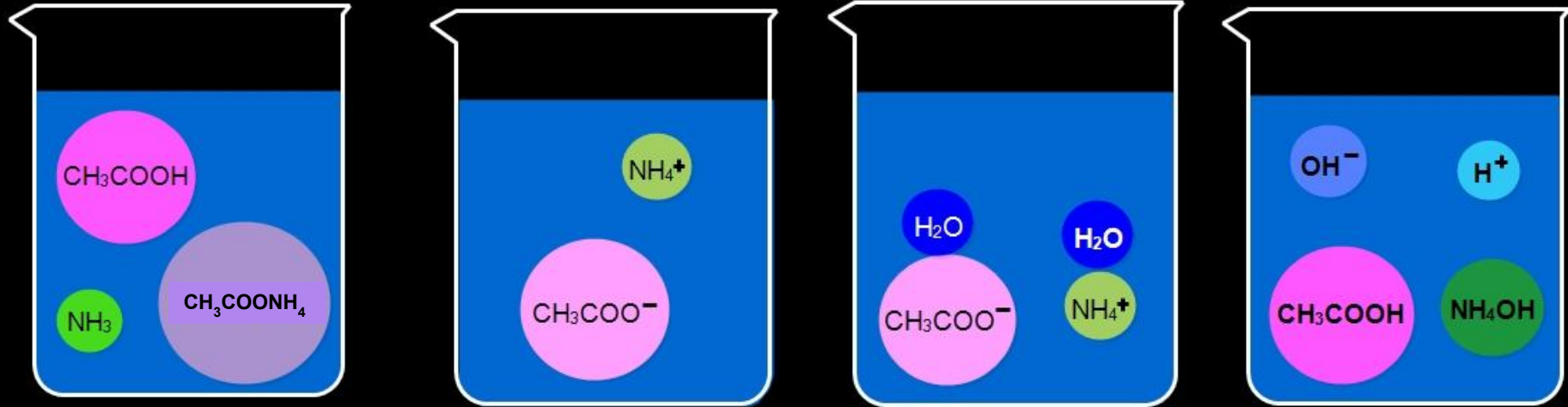
$$\% \text{ hydrolysis} = 100 \times h = 0.01\%$$

11C07.4

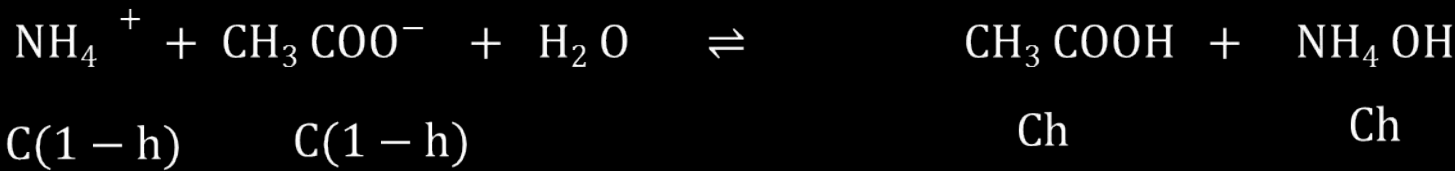
CV4

Hydrolysis of weak acid and weak base

# Hydrolysis of weak acid and weak base



# Hydrolysis of salt of weak acid and weak base



$$K_h = \frac{[\text{CH}_3\text{COOH}][\text{NH}_4\text{OH}]}{[\text{CH}_3\text{COO}^-][\text{NH}_4^+]}$$

$$K_h = \frac{Ch \times Ch}{C(1-h) \times C(1-h)} = h^2$$

$$= \frac{[\text{CH}_3\text{COOH}]}{[\text{CH}_3\text{COO}^-][\text{H}^+]} \times \frac{[\text{NH}_4\text{OH}]}{[\text{NH}_4^+][\text{OH}^-]} \times [\text{H}^+][\text{OH}^-]$$

$\frac{1}{K_a}$   $\frac{1}{K_b}$

$$= \frac{1}{K_a} \times \frac{1}{K_b} \times K_w$$

$$h^2 = \frac{K_w}{K_a \times K_b} = K_h$$

$$h = \sqrt{\frac{K_w}{K_a \times K_b}}$$

# Hydrolysis of salt of weak acid and weak base

We get  $H^+$  from dissociation of acid



$$K_a = \frac{C(1-h) \times [H^+]}{Ch}$$

$$= \frac{[H^+]}{h}$$

$$[H^+] = K_a \times h$$

$$= K_a \times \sqrt{\frac{K_w}{K_a \times K_b}}$$

$$[H^+] = \sqrt{\frac{K_w \times K_a}{K_b}}$$

$$pH = -\log \sqrt{\frac{K_w K_a}{K_b}}$$

$$pH = \frac{1}{2} \times (pK_w + pK_a - pK_b)$$

at 298 K

$$pH = 7 + \frac{1}{2} (pK_a - pK_b)$$

## Summary

- The salts of strong acid and weak base, weak acid and strong base, and weak acid and weak base undergo hydrolysis in aqueous solutions.
- Salts of strong acids and bases do not undergo hydrolysis, Hence pH of its solution will be 7.
- $\text{pH} = 7 - \frac{1}{2}(\text{pK}_b + \log C)$  for salt of strong acid and weak base
- $\text{pH} = 7 + \frac{1}{2}(\text{pK}_a + \log C)$  for salt of strong base and weak acid
- $\text{pH} = 7 + \frac{1}{2}(\text{pK}_a - \text{pK}_b)$  for salt of weak acid and weak base

## Reference Questions

NCERT In-text Questions: 25

11C07.4

PSV 3

Q. The  $pK_a$  of Formic acid and  $pK_b$  of Ammonium hydroxide are 3.74 and 4.75 respectively. Calculate the pH of ammonium formate solution.

Sol.

$$pH = 7 + \frac{1}{2} [pK_a - pK_b]$$

$$= 7 + \frac{1}{2} [3.74 - 4.75]$$

$$= 7 + \frac{1}{2} [-1.01]$$

$$= 7 - 0.505$$

$$= 6.495$$

11C07.5

Buffer Solution

# 11C07.5 – Buffer Solution

Learning objective

Introduction to Buffer

Acidic Buffer

Basic Buffer

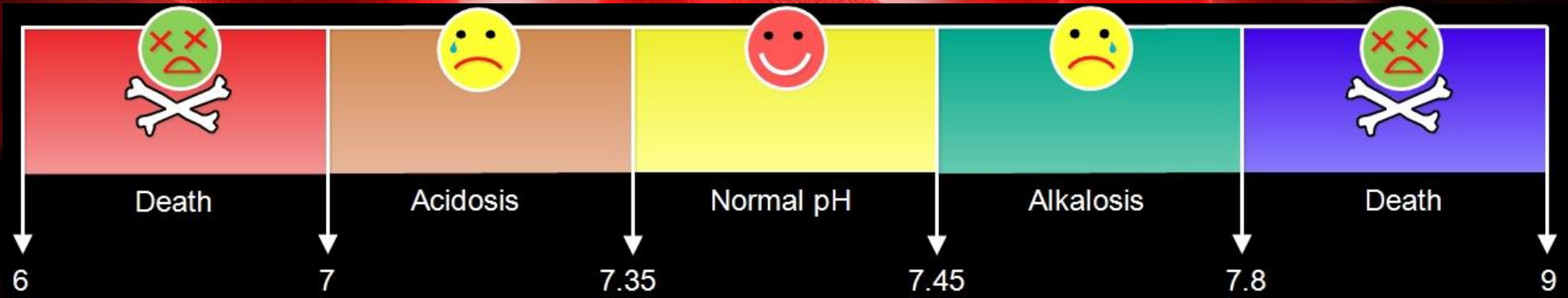
11C07.5

CV 1

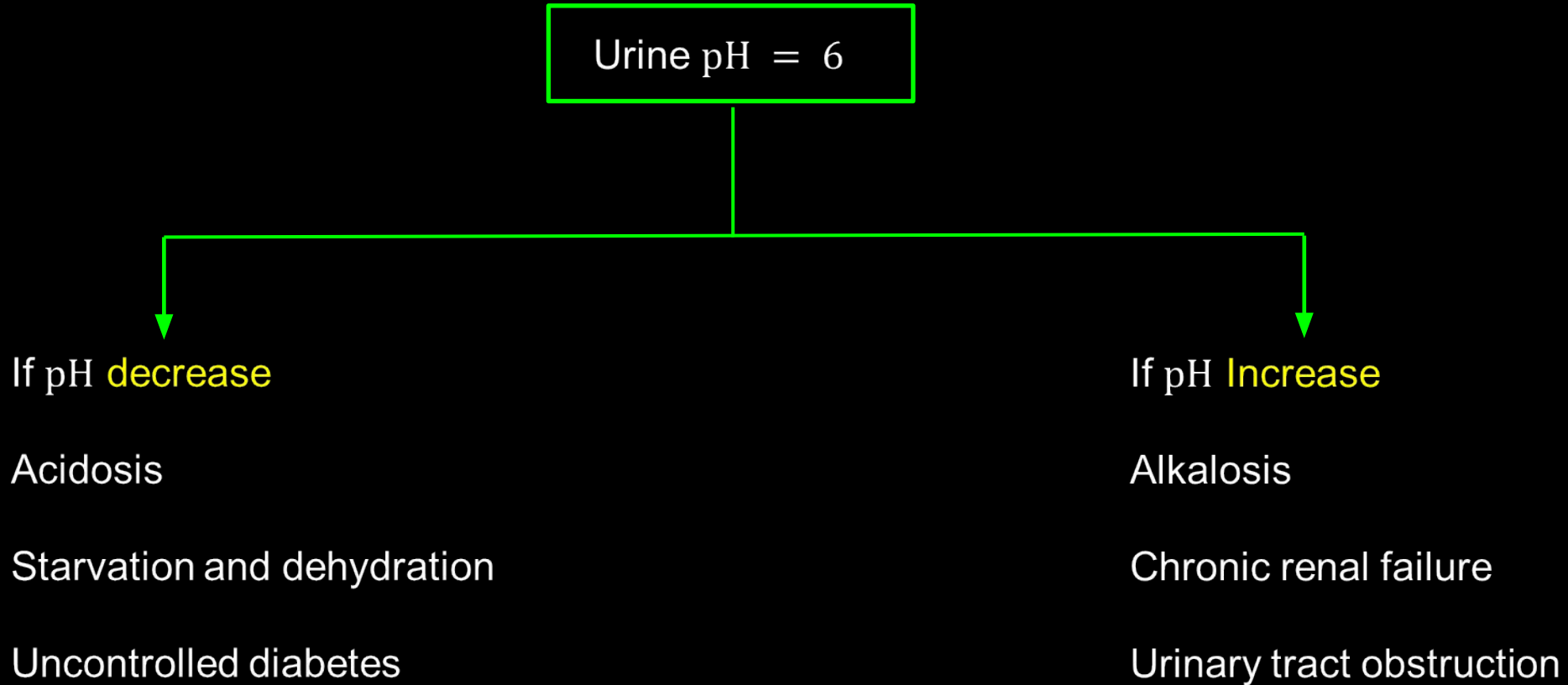
Introduction to Buffer

# Importance of Buffer Solution

pH Level of Blood



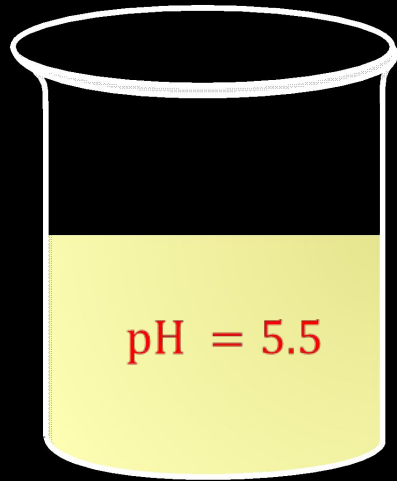
# Importance of Buffer Solution



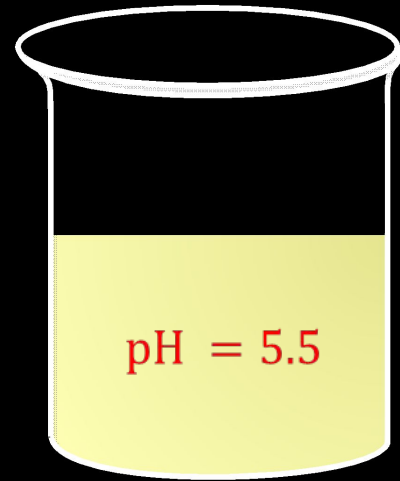
**Note:** Control of pH is important in many chemical and biochemical processes

# Buffer Solution

The solution which **resist change in pH** on dilution or with the addition of small amounts of acid or alkali are called Buffer solutions.

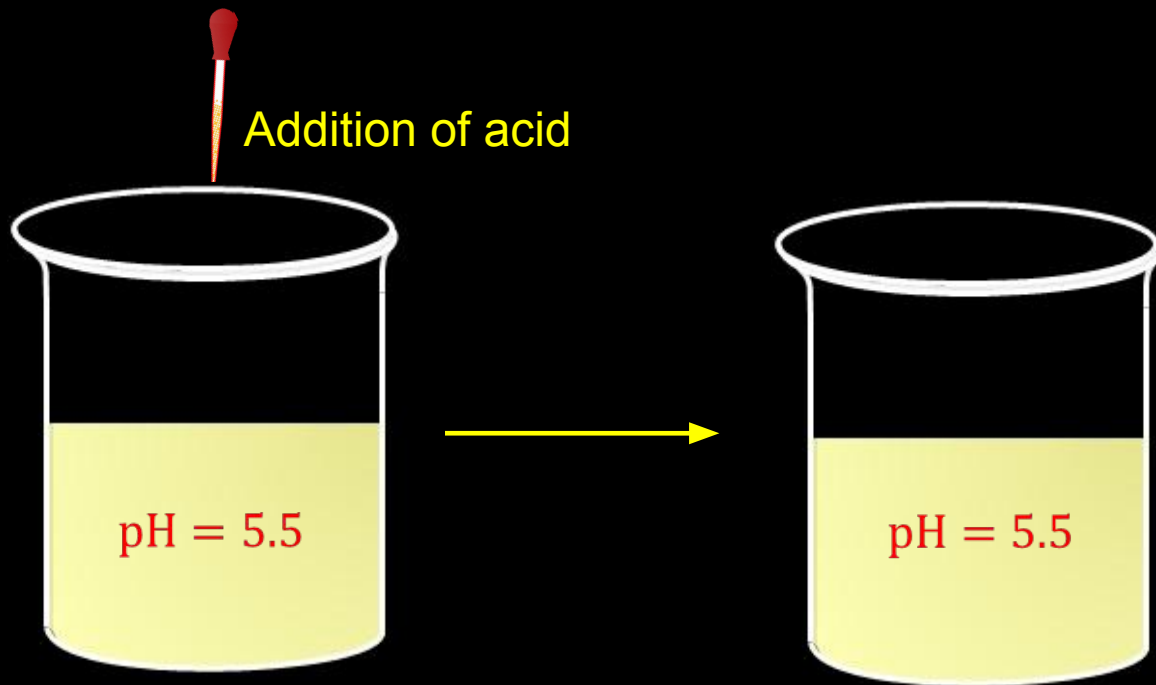


Addition of water



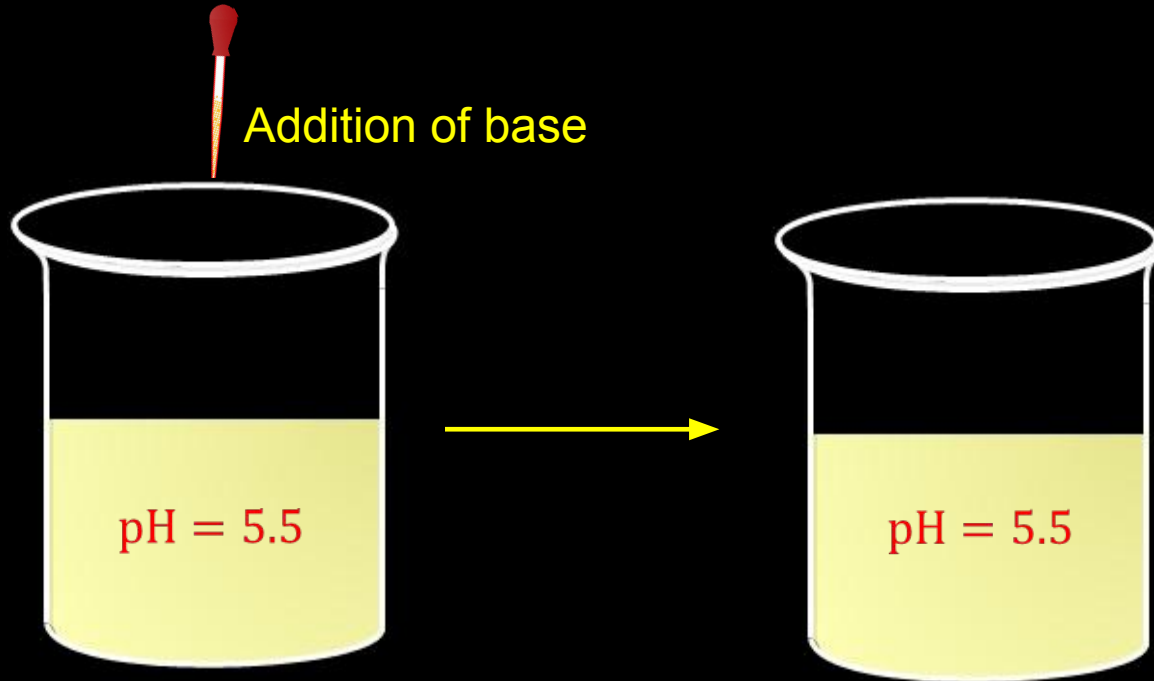
# Buffer Solution

The solution which **resist change in pH** on dilution or with the addition of small amounts of acid or alkali are called Buffer solutions.



# Buffer Solution

The solution which **resist change in pH** on dilution or with the addition of small amounts of acid or alkali are called Buffer solutions.



# Types of Buffer

## Types of Buffer

```
graph TD; A[Types of Buffer] --> B[Acidic Buffer]; A --> C[Basic Buffer]
```

### Acidic Buffer

Buffer solution which has  $\text{pH} < 7$

It is prepared by mixing **weak acid** and **its salt with a strong base**

**Example:**  $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$

### Basic Buffer

Buffer solution which has  $\text{pH} > 7$

It is prepared by mixing **weak base** and **its salt with a strong acid**

**Example:**  $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$

# Concept Test

Ready for challenge

Q. Can we obtain Buffer solution if we mix 100ml of 0.2 M  $\text{CH}_3\text{COOH}$  and 100ml of 0.1M NaOH.

Pause the video

(Time Duration: 02 Minute)

Q. Can we obtain Buffer solution if we mix 100ml of 0.2 M  $\text{CH}_3\text{COOH}$  and 100ml of 0.1M  $\text{NaOH}$ .

Sol.



Initial Conc. (20 mmol) (10 mmol)

Equilibrium Conc. (10 mmol) (0 mmol) (10 mmol)

Thus it is a **buffer solution**

11C07.5

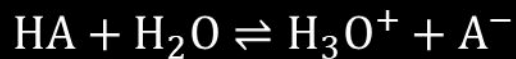
CV 2

Acidic Buffer

# Acidic Buffer

It is prepared by weak acid and its salt with a strong base

if the weak acid **HA ionises** in water,



$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$$

**Rearranging** the expression

$$[\text{H}_3\text{O}^+] = \frac{K_a[\text{HA}]}{[\text{A}^-]}$$

**Taking – ve logarithm** on both side

$$-\log[\text{H}_3\text{O}^+] = -\log K_a - \log \frac{[\text{HA}]}{[\text{A}^-]}$$

$$\text{pH} = \text{p}K_a - \log \frac{[\text{HA}]}{[\text{A}^-]}$$

$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

$$\text{pH} = \text{p}K_a + \log \frac{[\text{Cojugate Base}]}{[\text{Acid}]}$$

**Henderson Hasselbalch equation**

## Acidic Buffer

$$\text{pH} = \text{pK}_a + \log \frac{[\text{Conjugate Base}]}{[\text{Acid}]}$$

most of the conjugate base comes from **ionisation of salt** of the acid

$$\text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

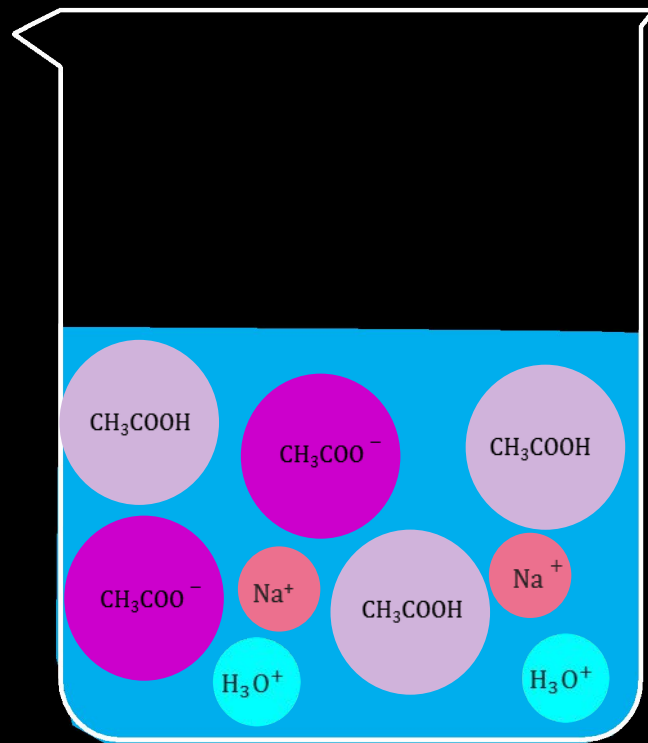
if  $[\text{Salt}] = [\text{Acid}]$  then,

$$\text{pH} = \text{pK}_a$$

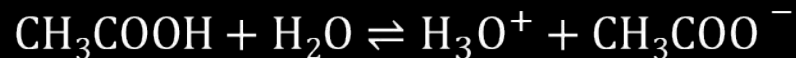
if we prepare buffer solution by mixing **Acetic acid** and **Sodium Acetate** in equal conc. then this solution has **pH = 4.76**

# Buffer Action of Acidic Buffer

we take example of buffer  $\text{CH}_3\text{COOH}$  and  $\text{CH}_3\text{COONa}$  then

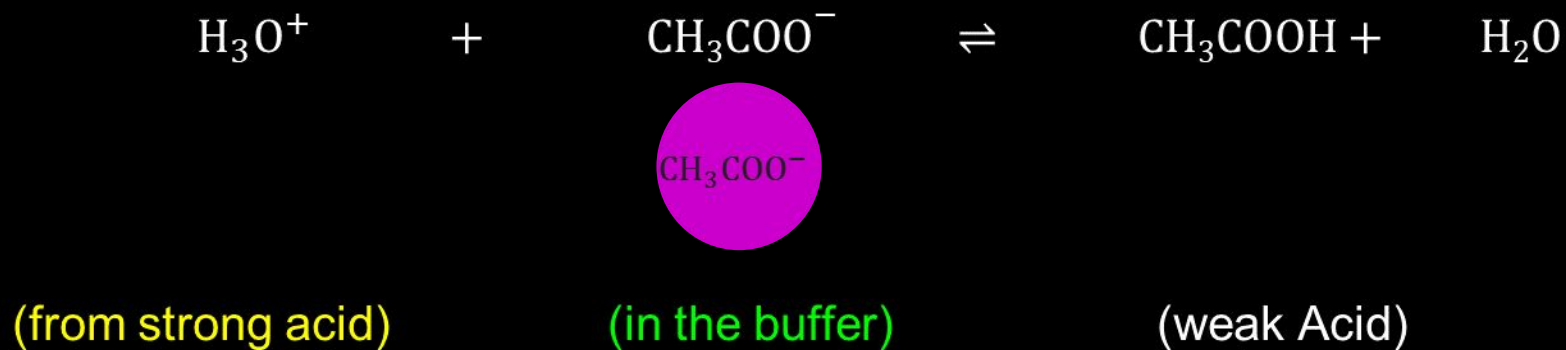


Buffer solution of  $\text{CH}_3\text{COOH}$   
and  $\text{CH}_3\text{COONa}$



# Buffer Action of Acidic Buffer

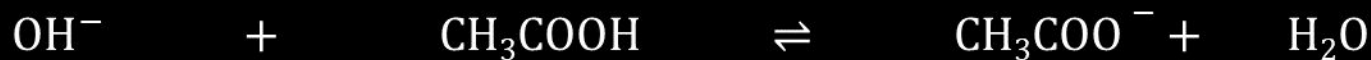
**Case- 1** If we add small amount of **strong acid** like HCl



Weak acid formed is which is less ionised so **pH** of buffer solution **not changed**

# Buffer Action of Acidic Buffer

**Case- 2** If we add small amount of **strong base** like NaOH



(from strong base)

(in the buffer)

$\text{CH}_3\text{COO}^-$  and  $\text{H}_2\text{O}$  formed so **pH** of buffer solution **not changed**

# Concept Test

Ready for challenge

Q. A buffer is prepared containing 1 M acetic acid and 0.8 M sodium acetate. What is its pH ? ( $pK_a$  for acetic acid is 4.752)

Pause the video

(Time Duration: 02 Minutes)

**Q.** A buffer is prepared containing 1 M acetic acid and 0.8 M sodium acetate. What is its pH ? ( $pK_a$  for acetic acid is 4.752)

**Sol.** From Henderson Hasselbalch equation

$$pH = pK_{a1} + \log \frac{[Salt]}{[Acid]}$$

$$pH = 4.752 + \log \frac{[0.8]}{[1.0]}$$

$$pH = 4.752 - 0.097$$

$$pH = 4.655$$

11C07.5

PSV 1

**Q.** Calculate the pH of buffer solution when 25ml of 0.2M acetic acid is mixed with 35ml of 0.1M NaOH (if pKa for acetic acid is 4.752)

**Sol.**

$$\text{Moles of CH}_3\text{COOH} = 0.2\text{M} \times 0.025\text{ L} = 0.005\text{ mole}$$

$$\text{Moles of NaOH} = 0.1 \times 0.035\text{ L} = 0.0035\text{ mole}$$



|             |           |            |
|-------------|-----------|------------|
| Initial mol | 0.005 mol | 0.0035 mol |
|-------------|-----------|------------|

|                    |                    |        |        |
|--------------------|--------------------|--------|--------|
| At Equilibrium mol | $(0.005 - 0.0035)$ | 0.0035 | 0.0035 |
|--------------------|--------------------|--------|--------|

$$= 0.00150$$

**Q.** Calculate the pH of buffer solution when 25ml of 0.2M acetic acid is mixed with 35ml of 0.1M NaOH (if pK<sub>a</sub> for acetic acid is 4.752)

**Sol.**

From Henderson Hasselbalch equation

$$\text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

$$\text{pH} = \text{pK}_a + \log \frac{[0.0035]}{[0.0015]}$$

$$\text{pH} = 4.752 + 0.368$$

$$\text{pH} = 5.120$$

11C07.5

CV 3

Basic Buffer

# Basic Buffer

It is prepared by weak base and its salt with a strong acid

If the weak base **BOH ionizes** in water



$$K_b = \frac{[\text{B}^+][\text{OH}^-]}{[\text{BOH}]}$$

**Rearranging** the expression

$$[\text{OH}^-] = \frac{K_b [\text{BOH}]}{[\text{B}^+]}$$

**Taking - ve logarithm** on both sides

$$-\log[\text{OH}^-] = -\log K_b - \log \frac{[\text{BOH}]}{[\text{B}^+]}$$

$$\text{pOH} = \text{p}K_b - \log \frac{[\text{BOH}]}{[\text{B}^+]}$$

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{B}^+]}{[\text{BOH}]}$$

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{Conjugate Acid}]}{[\text{Base}]}$$

**Henderson Hasselbalch equation**

## Basic Buffer

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{Conjugate Acid}]}{[\text{Base}]}$$

most of the conjugate acid comes from **ionisation of salt** of the acid

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{Salt}]}{[\text{Base}]}$$

we know,

$$\text{pH} = \text{pK}_w - \text{pOH}$$

$$\text{pH} = \text{pK}_w - \left( \text{pK}_b + \log \frac{[\text{Salt}]}{[\text{Base}]} \right)$$

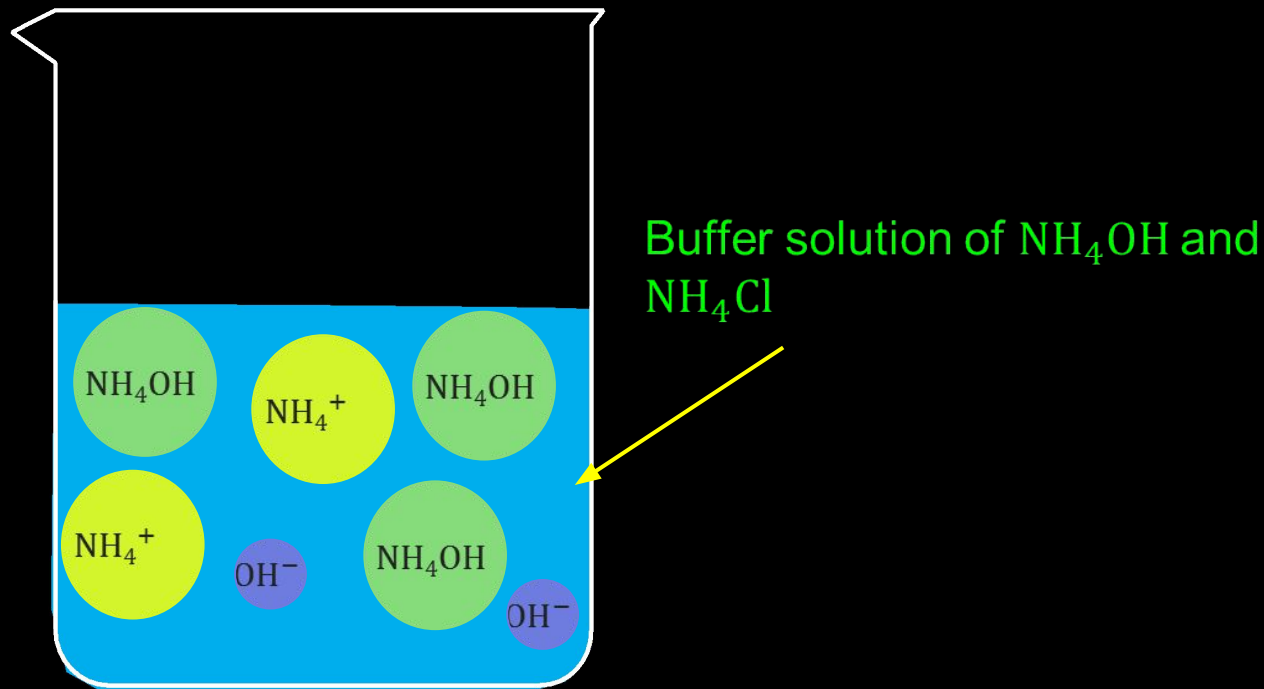
At 298 K,  $\text{pK}_w = 14$  so,

$$\text{pH} = 14 - \left( \text{pK}_b + \log \frac{[\text{Salt}]}{[\text{Base}]} \right)$$

If we prepare buffer solution by mixing **Ammonia solution** and **Ammonium chloride** in equal conc. then this solution has **pH = 9.25**

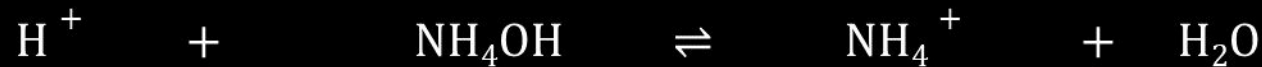
# Buffer Action of Basic Buffer

we take example of buffer  $\text{NH}_4\text{OH}$  and  $\text{NH}_4\text{Cl}$  then,



# Buffer Action of Basic Buffer

**Case- 1** If we add small amount of **strong acid** like HCl



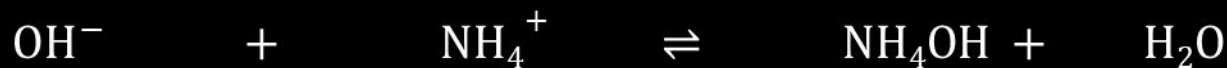
(from strong acid)

(in the buffer)

NH<sub>4</sub><sup>+</sup> and H<sub>2</sub>O formed so **pH** of buffer solution **not change**

# Buffer Action of Basic Buffer

**Case- 2** If we add small amount of **strong base** like NaOH



(from strong base)

(in the buffer)

Weak base formed is which is less ionised so **pH** of buffer solution **not changed**

## Summary

- The solution which resist change in pH on dilution or with the addition of small amounts of acid or alkali are called buffer solutions.
- Acidic buffer is prepared by weak acid and its salt with a strong base
- Henderson Hasselbalch equation for Acidic buffer Solution

$$\text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

- Basic buffer is prepared by weak base and its salt with a strong acid
- Henderson Hasselbalch equation for Buffer buffer Solution

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{Salt}]}{[\text{Base}]}$$

11C07.5

PSV 2

**Q.** Calculate the pH of buffer solution Consisting of 0.75M  $\text{NH}_4\text{Cl}$  and 1.25M  $\text{NH}_3$  ( $K_b$  for  $\text{NH}_3 = 1.8 \times 10^{-5}$  )

**Sol.**

$$\text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Base}]}$$

$$\text{pK}_b = -\log k_b$$

$$\text{pK}_b = -\log 1.8 \times 10^{-5}$$

$$\text{pK}_b = 4.7$$

$$\text{pK}_a + \text{pK}_b = 14$$

$$\text{pK}_a = 14 - 4.7$$

$$\text{pK}_a = 9.3$$

$$\text{pH} = 9.3 + \log \frac{[1.25]}{[0.75]}$$

$$\text{pH} = 9.3 + (\log 5 - \log 3)$$

$$\text{pH} = 9.3 + 0.2219$$

$$\text{pH} = 9.5219$$

11C07.6

Solubility product

# 11C07.6 – Solubility product

Learning objective

Solubility Equilibria

Writing Expression for  $K_{sp}$

Effect of Common ion on Solubility

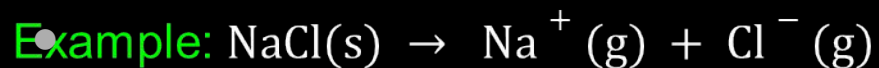
11C07.6

CV 1

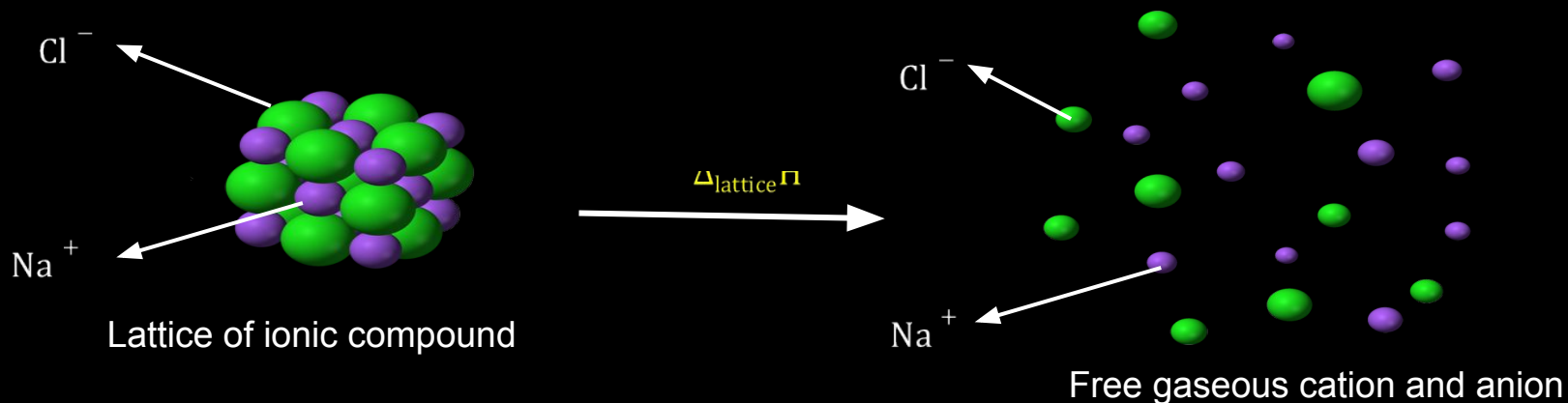
Solubility Equilibria

# Lattice Energy

Energy required to **break 1 mole** of ionic compound in solid state to form **free gaseous ions**.

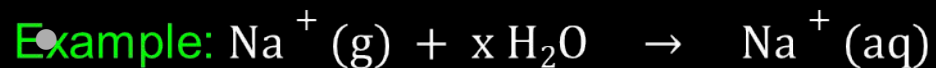


$$\Delta_{\text{lattice}}H = +787 \text{ kJ/mol}$$

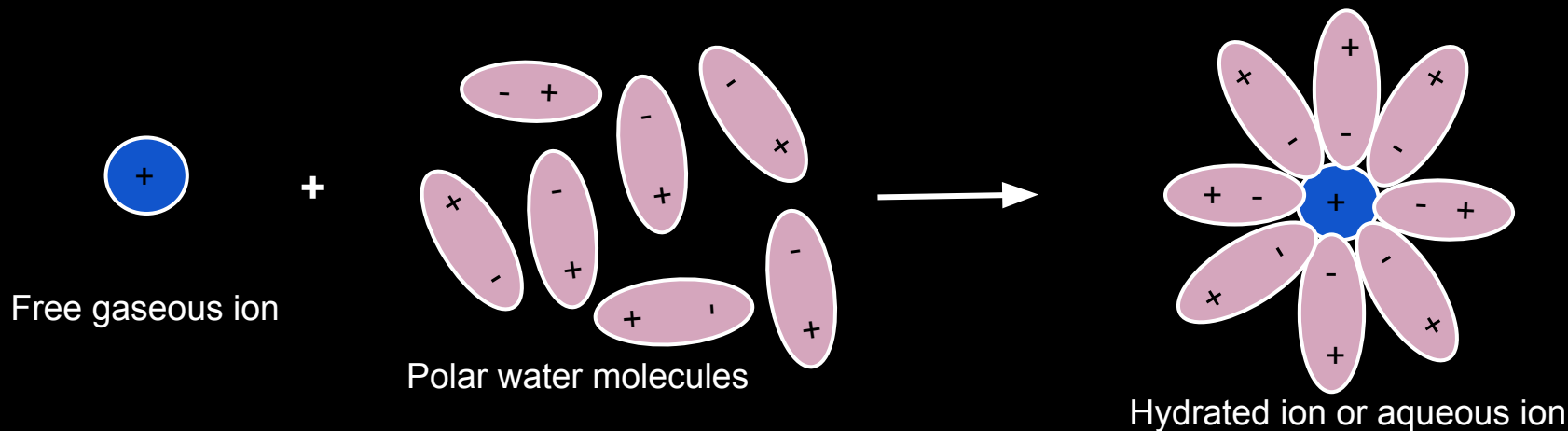


# Solvation Enthalpy/ Hydration Enthalpy

Enthalpy change when **1 mole of a substance or ion** is **completely hydrated** by water.



$$\Delta_{\text{hyd}}H^\circ = -406 \text{ kJ/mol}$$



# Solubility

The maximum amount of solute that dissolve in one unit volume of the solvent at a given temperature and pressure.

Case-1                  Solvation enthalpy > Lattice Enthalpy                  (Salt dissolve)

Case-2                  Lattice Enthalpy > Solvation Enthalpy                  (Salt does not dissolve)

.

# Classification of Salt on the basis of Solubility

|              |                   |                                                  |
|--------------|-------------------|--------------------------------------------------|
| Category I   | Soluble           | Solubility $> 0.1\text{M}$                       |
| Category II  | Slightly Soluble  | $0.01\text{M} < \text{Solubility} < 0.1\text{M}$ |
| Category III | Sparingly Soluble | Solubility $< 0.01\text{M}$                      |

# Concept Test

Ready for challenge

Q. A Chemist adds 10 g of  $\text{PbCl}_2$  to 50 ml of water at  $25^\circ\text{C}$  and finds that only 0.22 g of the solid goes in to solution. Calculate the solubility in g/L and mol/L.  
(molar mass of  $\text{PbCl}_2 = 278.1 \text{ g/mol}$ )

(Time Duration: 02 Minute)

Q. A Chemist adds 10 g of  $\text{PbCl}_2$  to 50 ml of water at  $25^\circ\text{C}$  and finds that only 0.22 g of the solid goes in to solution. Calculate the solubility in g/L and mol/L.  
(molar mass of  $\text{PbCl}_2 = 278.1 \text{ g/mol}$ )

$$\text{volume of solution in Litre} = \frac{50}{1000} = 0.05\text{L}$$

$$\text{Solubility in } \frac{\text{g}}{\text{L}} = \frac{0.22\text{g}}{0.05\text{L}} = 4.4 \frac{\text{g}}{\text{L}}$$

$$\text{Mole of } \text{PbCl}_2 = \frac{0.22\text{g}}{278.1\text{g/mol}} = 0.00079\text{mol}$$

$$\text{Solubility in } \frac{\text{mol}}{\text{L}} = \frac{0.00079\text{mol}}{0.05\text{L}} = 0.0158 \frac{\text{mol}}{\text{L}}$$

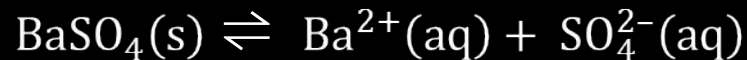
11C07.6

CV 2

Solubility Product  $K_{sp}$

# Solubility product

A Solubility equilibrium exists when a chemical compound in the solid state is in chemical equilibrium with a solution of that compound



|                  |   |   |   |
|------------------|---|---|---|
| Initial conc.    | s | 0 | 0 |
| Equilibrium conc |   | s | s |

$$K = \frac{[\text{Ba}^{2+}][\text{SO}_4^{2-}]}{[\text{BaSO}_4]}$$

$$K[\text{BaSO}_4] = [\text{Ba}^{2+}][\text{SO}_4^{2+}]$$

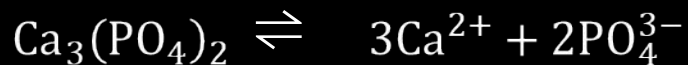
$$K_{\text{sp}} = [\text{Ba}^{2+}][\text{SO}_4^{2+}]$$

$$1.1 \times 10^{-10} = s \times s = s^2$$

$$s = 1.05 \times 10^{-5} \text{ mol L}^{-1}$$

# Solubility product

Example-



Initial Conc.

$s$

$0$

$0$

Equilibrium conc

$0$

$3s$

$2s$

$$K_{sp} = [\text{Ca}^{2+}]^3 [\text{PO}_4^{3-}]^2$$

$$[\text{Ca}^{2+}] = 3s \quad [\text{PO}_4^{3-}] = 2s$$

$$K_{sp} = (3s)^3 \times (2s)^2$$

$$K_{sp} = 27s^3 \times 4s^2$$

$$K_{sp} = 108s^5$$

# Concept Test

Ready for challenge

Q. The solubility product constant of  $\text{Ag}_2\text{CrO}_4$  and  $\text{AgBr}$  are  $1.1 \times 10^{-12}$  and  $5.0 \times 10^{-13}$  respectively. Calculate the ratio of the molarities of their saturated solutions.

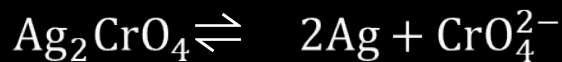
Pause the video

(Time Duration: 02 Minute)

**Q.** The solubility product constant of  $\text{Ag}_2\text{CrO}_4$  and  $\text{AgBr}$  are  $1.1 \times 10^{-12}$  and  $5.0 \times 10^{-13}$  respectively. Calculate the ratio of the molarities of their saturated solutions.

**Sol.**

For  $\text{Ag}_2\text{CrO}_4$

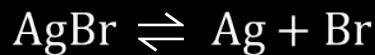


$$K_{\text{sp}} = (2s)^2 \times (s) = 4s^3$$

$$4s^3 = 1.1 \times 10^{-12}$$

$$s = 6.5 \times 10^{-5}\text{M}$$

For  $\text{AgBr}$



$$K_{\text{sp}} = (s) \times (s) = s^2$$

$$s^2 = 5.0 \times 10^{-13}$$

$$s = 7.07 \times 10^{-7}\text{M}$$

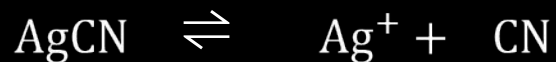
$$\frac{s \text{ for } \text{Ag}_2\text{CrO}_4}{s \text{ for } \text{AgBr}} = \frac{6.5 \times 10^{-5}}{7.07 \times 10^{-7}} = 91.9$$

11C07.6

PSV 1

**Q.** The values of  $K_{sp}$  of two sparingly soluble salts  $\text{Ni}(\text{OH})_2$  and  $\text{AgCN}$  are  $2.0 \times 10^{-15}$  and  $6 \times 10^{-17}$  respectively. Which salt is more soluble? Explain.

**Sol.**



$$K_{sp} = [\text{Ag}^+][\text{CN}^-] = 6 \times 10^{-17} = s^2$$

$$s_{\text{AgCN}} = 7.8 \times 10^{-9}$$



$$K_{sp} = [\text{Ni}^{2+}][\text{OH}^-]^2 = (s)(2s)^2 = 4s^3$$

$$K_{sp} = 4s^3 = 2 \times 10^{-15}$$

$$s_{\text{Ni}(\text{OH})_2} = 7.9 \times 10^{-6}$$

$$s_{\text{Ni}(\text{OH})_2} > s_{\text{AgCN}}$$

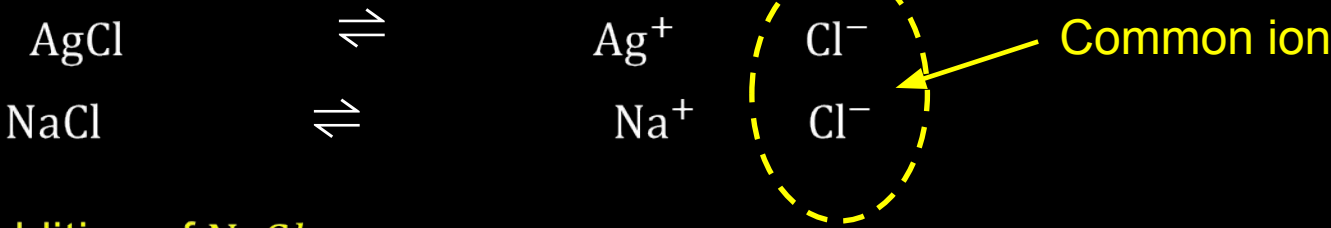
11C07.6

CV 3

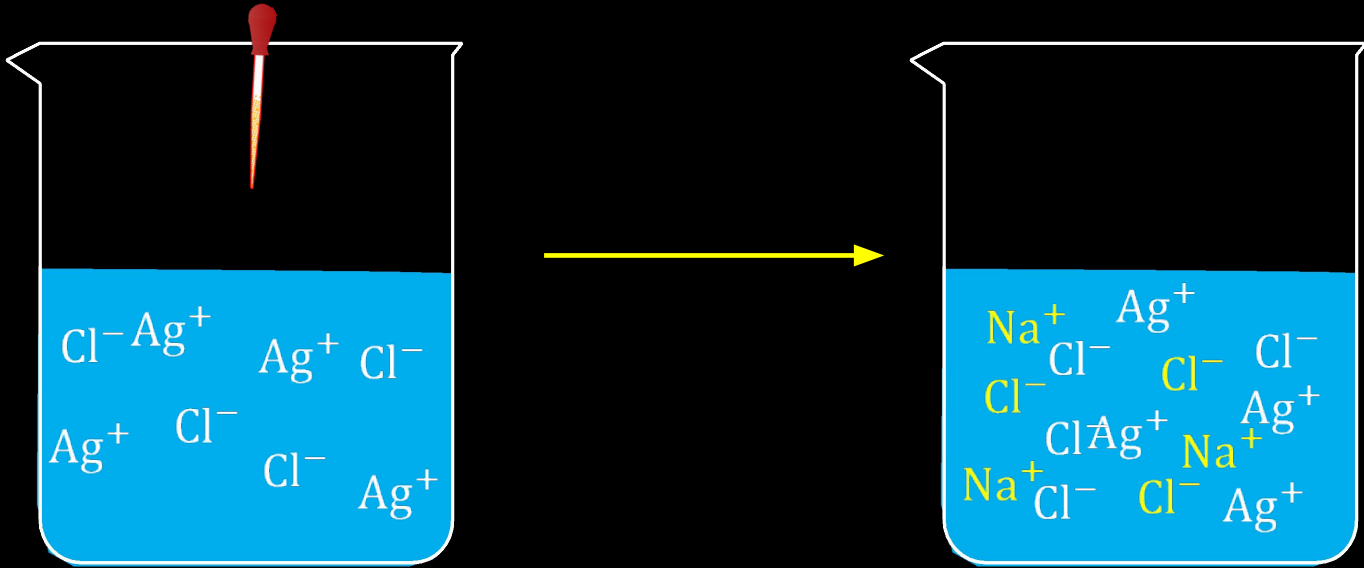
Effect of common ion on Solubility

# Effect of Common Ion

According to Le Chateliers principle



Addition of NaCl



# Condition for precipitation

$K_{sp}$  = Solubility product at Equilibrium Concentration

$Q_{sp}$  = Ionic product at any Concentration

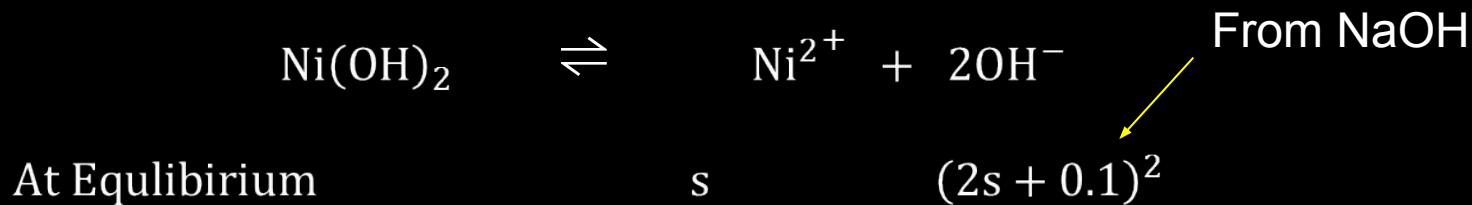
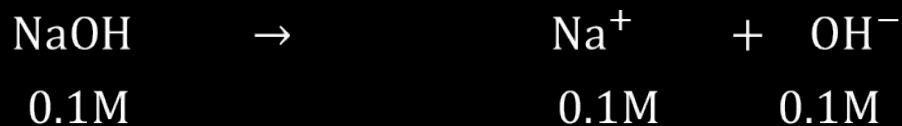
|  |                         |
|--|-------------------------|
|  | precipitate will form   |
|  | more salt will dissolve |
|  | At Equilibrium          |

11C07.6

PSV 2

**Q.** Calculate the molar solubility of  $\text{Ni(OH)}_2$  in  $0.10\text{M NaOH}$  The ionic product of  $\text{Ni(OH)}_2$  is  $2.0 \times 10^{-15}$

**Sol.**



$$K_{sp} = (s) \times (2s + 0.1)^2$$

As  $K_{sp}$  small,  $2s \ll 0.10$

$$\text{thus } (0.10 + 2s) \cong 0.10$$

$$2.0 \times 10^{-15} = s \times (0.10)^2$$

$$s = 2.0 \times 10^{-13}$$

## Summary

- Hydration enthalpy > Lattice enthalpy (Salt dissolve)
- Lattice enthalpy > Hydration enthalpy (Salt does not dissolve)

- Expression of Solubility product



$$K_{sp} = x^x y^y \times s^{x+y}$$

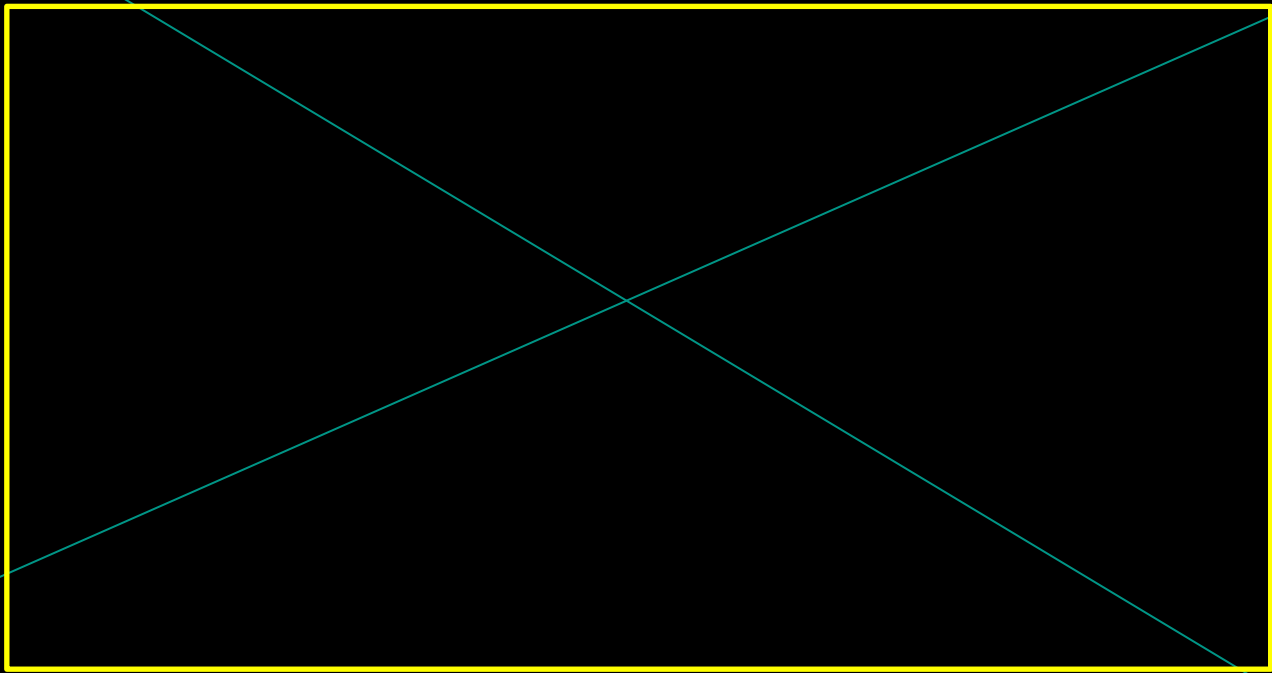
- Condition for precipitation  $K_{sp} < Q_{sp}$

## Reference Question

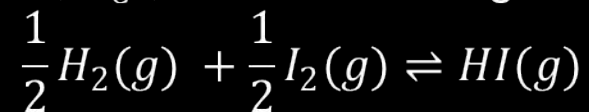
NCERT intext Question : 7.26, 7.27, 7.28

NCERT Exercise Question : 7.67, 7.68, 7.69, 7.70, 7.71, 7.72, 7.73

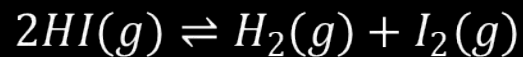
Workbook Question : 22

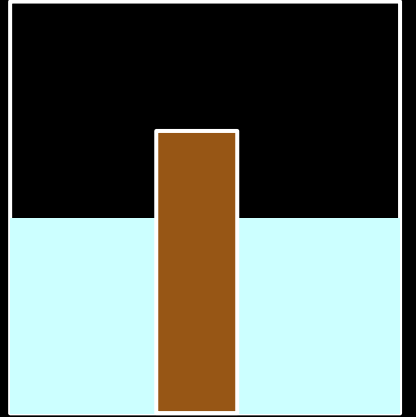
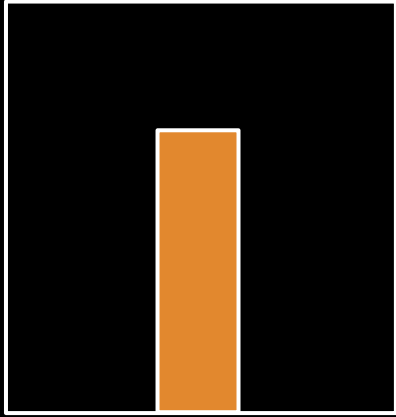
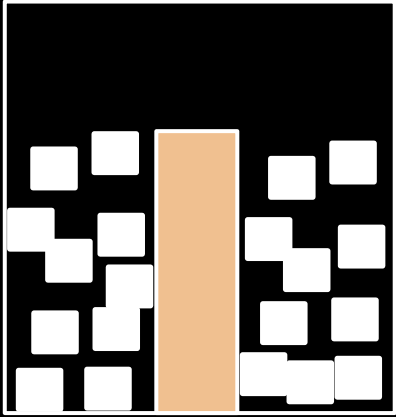


At 500 K, equilibrium constant,  $K_c$  , for the following reaction is 5.

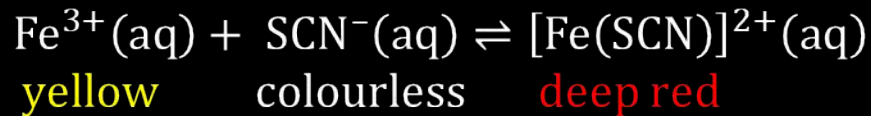


What would be the equilibrium constant  $K_c$  for the reaction

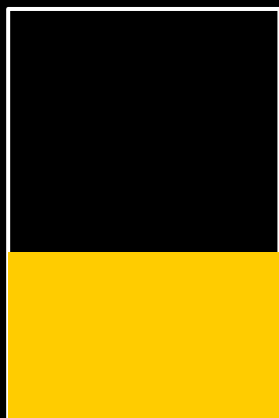




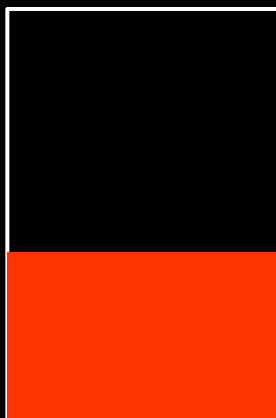
## Changing the Concentrations of Reactants and Products



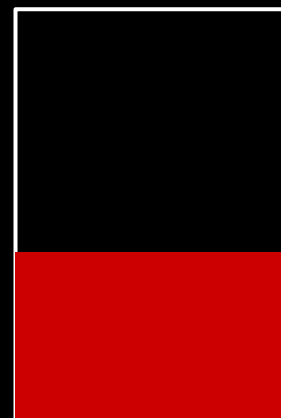
$$K_c = \frac{[\text{Fe}(\text{SCN})]^{2+}}{[\text{Fe}^{3+}][\text{SCN}^{-}]}$$

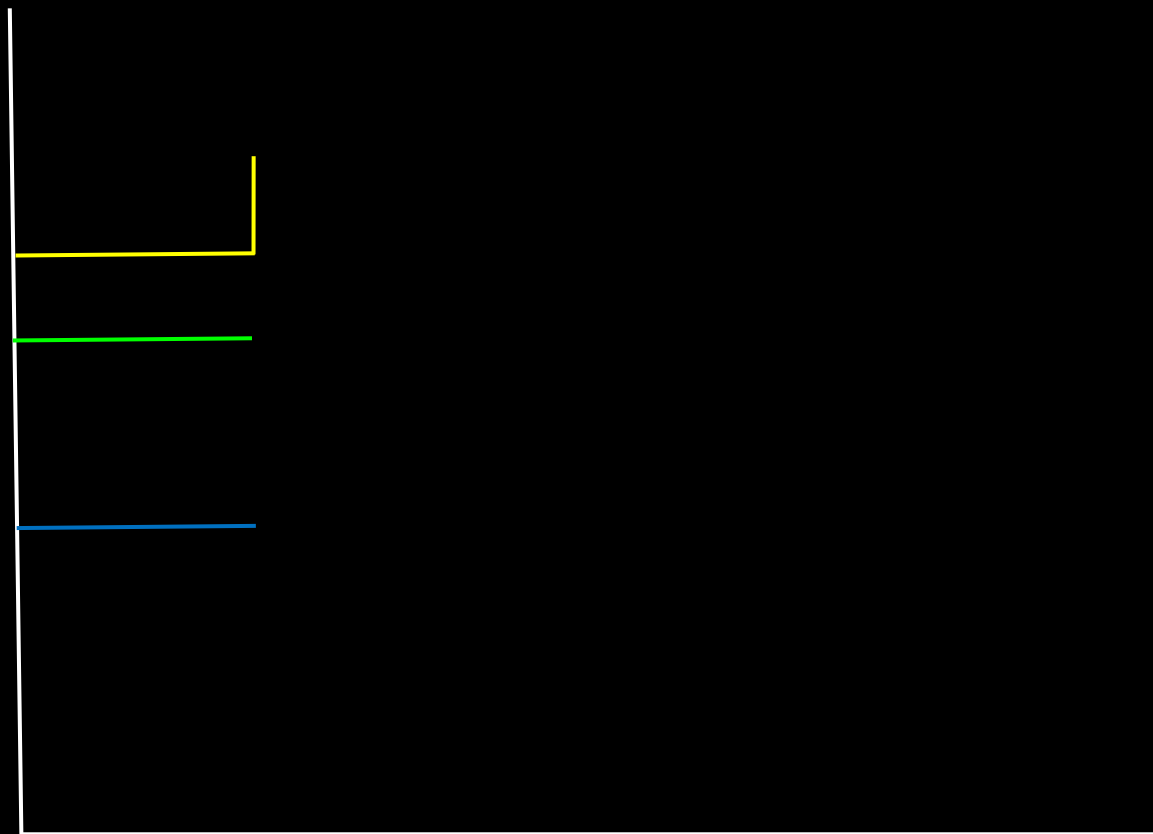


Addition of  
 $[\text{Fe}(\text{SCN})]^{2+}$   
←  
Backward  
Direction

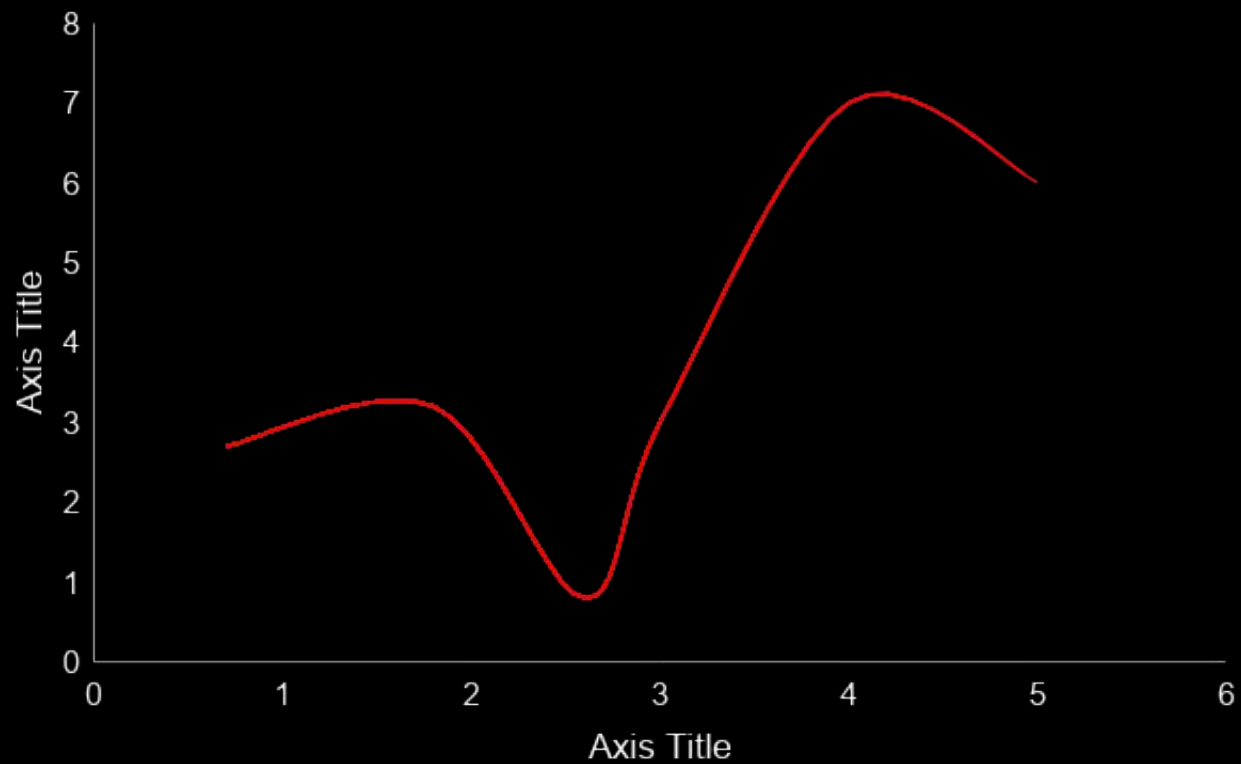


Addition of  
 $[\text{Fe}^{3+}]$  or  $[\text{SCN}^{-}]$   
→  
Forward  
Direction



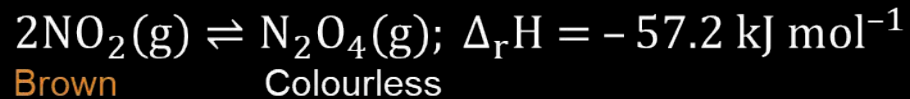


# Concentration

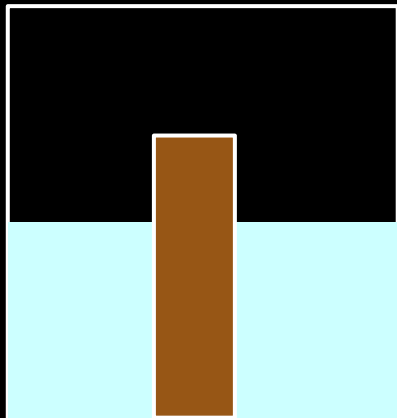




# Changing the Temperature

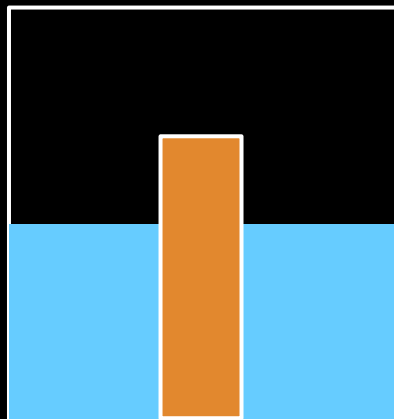


Water at 363 K



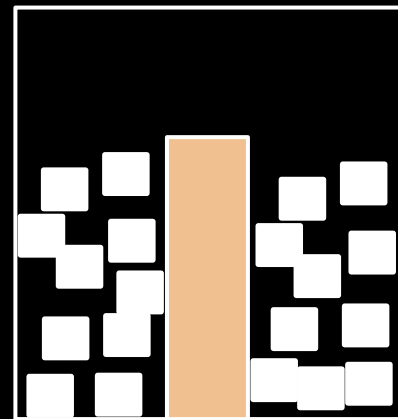
Backward  
Direction

Water at room  
temperature 270 K

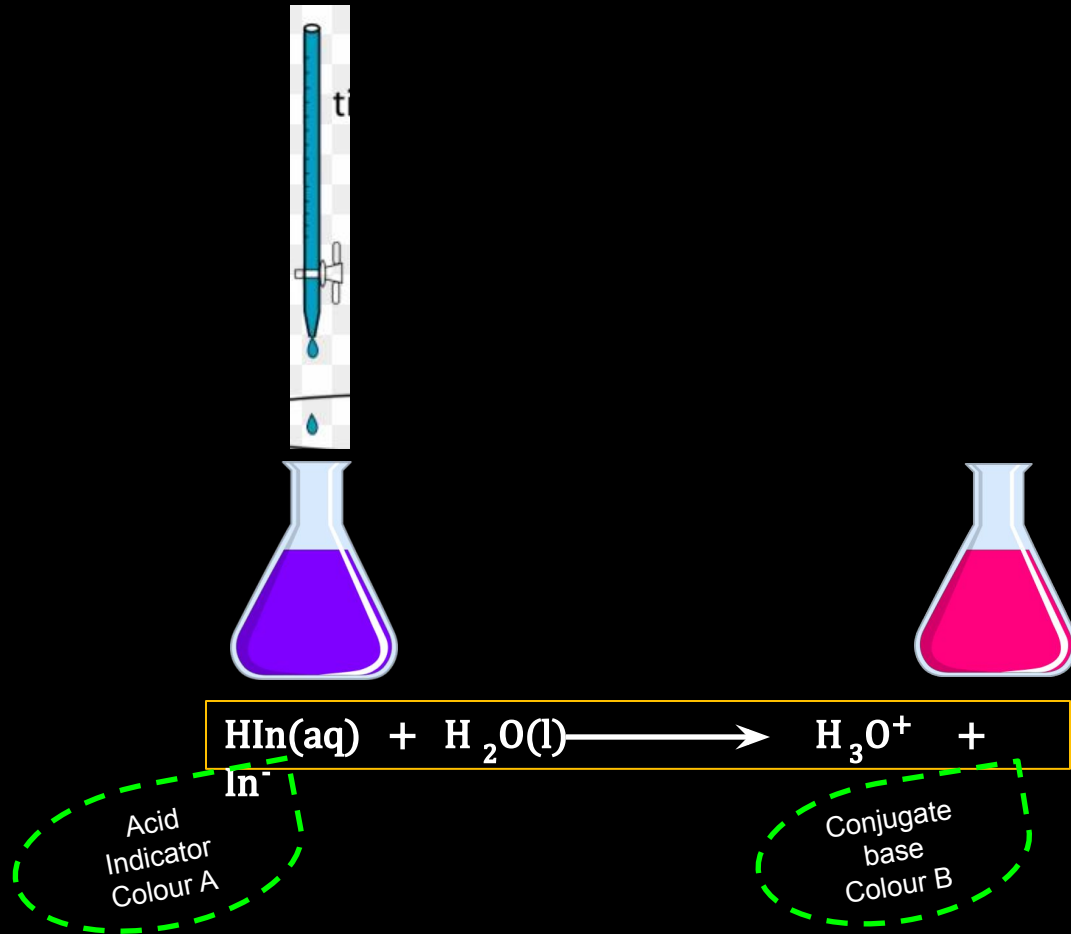


Forward  
Direction

Freezing mixture  
at 270 K



# Ionization of Acids



# Arrhenius Concept of Acids and Bases

Bases Dissociates in water to give hydroxyl ions ( $\text{OH}^-$ )

